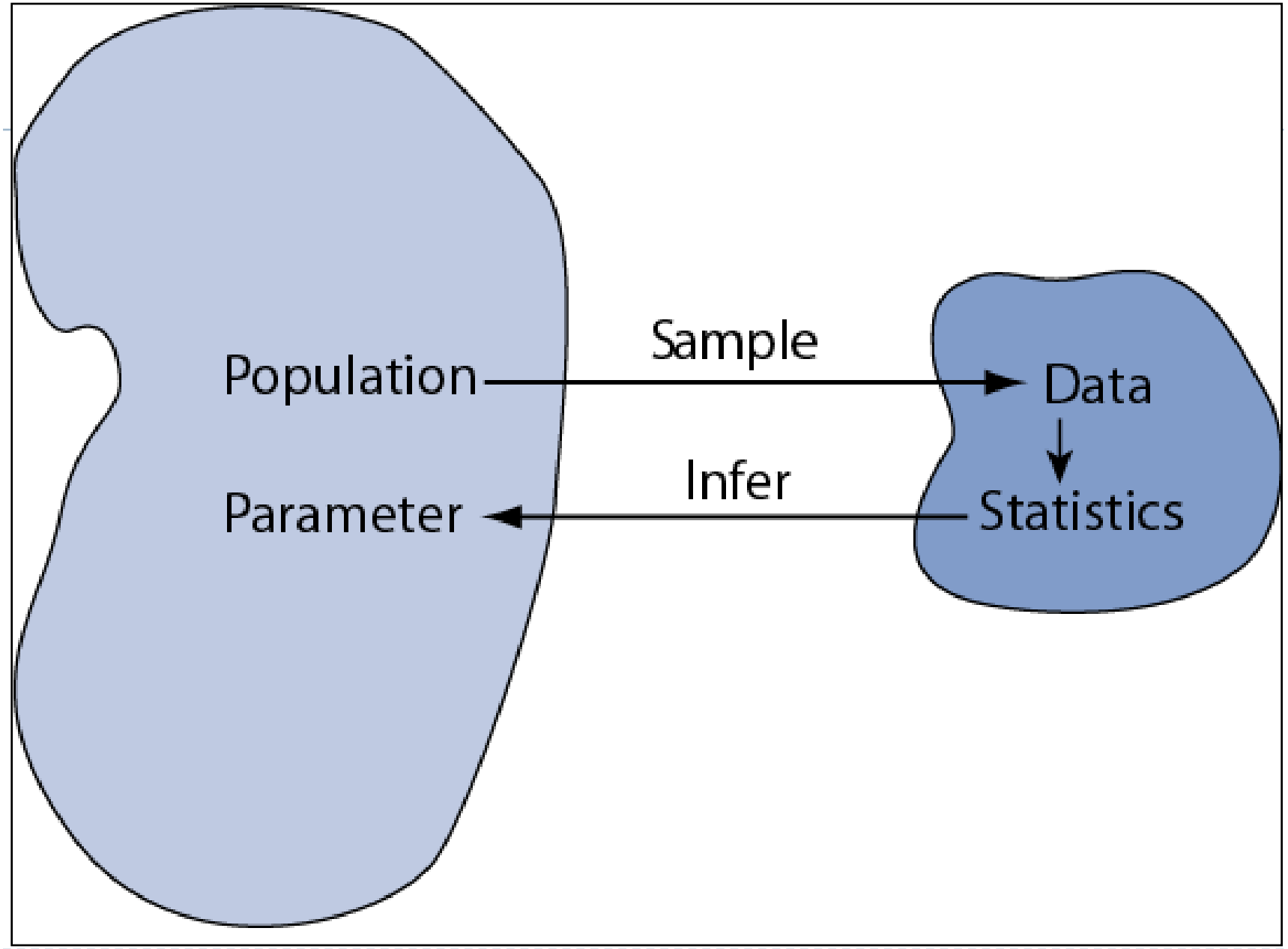
Statistics

NOU

Terminologies

- **Population** ≡ all possible values
- **Sample** ≡ a portion of the population
- **Statistical inference** ≡ generalizing from a sample to a population with calculated degree of certainty
- Two forms of statistical inference
  - **Hypothesis testing**
  - **Estimation**
- **Parameter** ≡ a characteristic of population, e.g., population mean μ
- **Statistic** ≡ calculated from data in the sample, e.g., sample mean ($\bar{x}$)




Population
Sample
Data
Infer
Parameter
Statistics

Sampling distributions

- Statistical inference draws conclusions about population on the basis of data.
- The data are summarized by statistics such as the sample mean and the sample standard deviation.
- When the data are produced by **random sampling** or **randomized experimentation**, a statistic is a random variable that obeys the laws of probability theory.
- The **link between probability and data** is formed by the sampling distributions of statistics.
- A sampling distribution shows how a statistic would vary in repeated data production.
- *"Sampling distribution is a probability distribution that determines probabilities of the possible values of a sample statistic."* (Agresti & Finlay 1997)



Sampling distributions

- Each statistic has a sampling distribution. A sampling distribution is simply **a type** of probability distribution.
- A sampling distribution of statistic based on *n* observations is the probability distribution for that statistic resulting from repeatedly taking samples of size *n*, each time calculating the statistic value.

https://www.khanacademy.org/math/ap-statistics/sampling-distribution-ap/what-is-sampling-distribution/v/introduction-to-sampling-distributions



Sampling distributions of sample means

- Select an simple random sample of size *n* from population, and measure a variable *X* on each individual in the sample.
- The data consist of observations on *n* random variables $X_1, X_2, \ldots, X_n$.
- A single X_i is a measurement on one individual selected at random from the population and therefore X_i is a random variable with probability distribution equaling the population distribution of variable X.
- If the population is large relatively to the sample, we can consider $X_1, X_2, \ldots, X_n$ to be independent random variables each having the same probability distribution.

The sample mean of an simple random sample of size n is

$$\bar{X} = \frac{X_1 + X_2 + \cdots + X_n}{n}.$$



Sampling distributions of sample means

If the population variable X has a population mean μ, the μ is also mean of each observation X_i. Therefore, by the addition rule for means of random variables,

$$\begin{aligned}
\mu_{\bar{X}} = E(\bar{X}) &= E\left(\frac{X_1 + X_2 + \cdots + X_n}{n}\right) \\
&= \frac{E(X_1 + X_2 + \cdots + X_n)}{n} \\
&= \frac{E(X_1) + E(X_2) + \cdots + E(X_n)}{n} \\
&= \frac{\mu_{X_1} + \mu_{X_2} + \cdots + \mu_{X_n}}{n} \\
&= \frac{\mu + \mu + \cdots + \mu}{n} \\
&= \mu.
\end{aligned}$$

That is, the mean of $\bar{X}$ is the same as the population mean μ of the variable X.



Sampling distributions of sample means

Furthermore, based on the addition rule for variances of independent random variables, X has the variance

$$\begin{aligned}\sigma_{\bar{X}}^2 &= \frac{\sigma_{X_1}^2 + \sigma_{X_2}^2 + \cdots + \sigma_{X_n}^2}{n^2} \\ &= \frac{\sigma^2 + \sigma^2 + \cdots + \sigma^2}{n^2} \\ &= \frac{\sigma^2}{n},\end{aligned}$$

and hence the standard deviation of $\bar{X}$ is

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}.$$

The standard deviation of $\bar{X}$ is also called the **standard error** of $\bar{X}$.



Sampling distributions of sample means

KEY FACT 7.1 (Mean and standard error of $\bar{X}$). *For a random sample of size n from a population having mean μ and standard deviation σ, the sampling distribution of the sample mean $\bar{X}$ has mean $\mu_{\bar{X}} = \mu$ and standard deviation, i.e., standard error $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$.* (Moore & McCabe, 1998)

The mean and standard error of $\bar{X}$ shows that the sample mean $\bar{X}$ tends to be closer to the population mean μ for larger values of n, since the sampling distribution becomes less spead about μ. This agrees with our intuition that larger samples provide more precise **estimates** of population characteristics.

KEY FACT 7.2 (Distribution of sample mean). *Suppose a variable X of a population is normally distributed with mean μ and standard deviation σ. Then, for samples of size n, the sample mean $\bar{X}$ is also normally distributed and has mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$. That is, if $X \sim N(\mu, \sigma)$, then $\bar{X} \sim N(\mu, \frac{\sigma}{\sqrt{n}})$.* (Weiss, 1999)



Sampling distributions of sample means

When sampling from nonnormal population, the distribution of $\bar{X}$ depends on what is the population distribution of the variable X. A surprising result, known as the **central limit theorem** states that when the sample size n is large, the probability distribution of the sample mean $\bar{X}$ is approximately normal, regardless of the shape of the population distribution.

KEY FACT 7.3 (Central limit theorem). *Whatever is the population distribution of the variable X, the probability distribution of the sample mean $\bar{X}$ is approximately normal when n is large. That is, when n is large, then*

$$\bar{X} \text{ approximately } N\left(\mu, \frac{\sigma}{\sqrt{n}}\right).$$

(Johnson & Bhattacharyya 1992)



Estimation

- Statistical inference uses sample data to form two types of estimators of parameters.
- A point estimate consists of a single number, calculated from the data, that is the best single guess for the unknown parameter.
- A interval estimate consists of a range of numbers around the point estimate, within which the parameter is believed to fall.



Point estimation

DEFINITION 8.1. *A point estimator of a unknown population parameter is a statistic that estimates the value of that parameter. A point estimate of a parameter is the value of a statistic that is used to estimate the parameter.* (Agresti & Finlay, 1997 and Weiss, 1999)

For instance, to estimate a population mean μ, perhaps the most intuitive point estimator is the sample mean:

$$\bar{X} = \frac{X_1 + X_2 + \cdots + X_n}{n}.$$

Once the observed values $x_1, x_2, \ldots, x_n$ of the random variables X_i are available, we can actually calculate the observed value of the sample mean $\bar{x}$, which is called a point estimate of μ.



- A good point estimator of a parameter is one with *sampling distribution that is centered around parameter, and has small standard error as possible*.
- A point estimator is called unbiased if its sampling distribution centers around the parameter in the sense that the *parameter is the mean of the distribution*.

For example, the mean of the sampling distribution of the sample mean $\bar{X}$ equals μ. Thus, $\bar{X}$ is an unbiased estimator of the population mean μ.



Point estimators of the population mean and standard deviation

The sample mean $\bar{X}$ is the obvious point estimator of a population mean μ. In fact, $\bar{X}$ is unbiased, and it is relatively efficient for most population distributions. It is the point estimator, denoted by $\hat{\mu}$, used in this text:

$$\hat{\mu} = \bar{X} = \frac{X_1 + X_2 + \cdots + X_n}{n}.$$

Moreover, the sample standard deviation s is the most popular point estimate of the population standard deviation σ. That is,

$$\hat{\sigma} = s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}.$$



Confidence interval

DEFINITION 8.2 (Confidence interval). *A confidence interval for a parameter is a range of numbers within which the parameter is believed to fall. The probability that the confidence interval contains the parameter is called the confidence coefficient. This is a chosen number close to 1, such as 0.95 or 0.99.* (Agresti & Finlay, 1997)

Confidence interval for μ when σ known

We first confine our attention to the construction of a confidence interval for a population mean μ assuming that the population variable X is *normally distributed* and its the standard deviation σ is *known*.

$$\bar{X} \sim N(\mu, \tfrac{\sigma}{\sqrt{n}})$$



$$P(\mu - 1.96\frac{\sigma}{\sqrt{n}} < \bar{X} < \mu + 1.96\frac{\sigma}{\sqrt{n}}) = 0.95.$$

Now the relation

$$\mu - 1.96\frac{\sigma}{\sqrt{n}} < \bar{X} \quad \text{equals} \quad \mu < \bar{X} + 1.96\frac{\sigma}{\sqrt{n}}$$

and

$$\bar{X} < \mu + 1.96\frac{\sigma}{\sqrt{n}} \quad \text{equals} \quad \bar{X} - 1.96\frac{\sigma}{\sqrt{n}} < \mu.$$

Hence the probability statement

$$P(\mu - 1.96\frac{\sigma}{\sqrt{n}} < \bar{X} < \mu + 1.96\frac{\sigma}{\sqrt{n}}) = 0.95$$

can also be expressed as

$$P(\bar{X} - 1.96\frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + 1.96\frac{\sigma}{\sqrt{n}}) = 0.95.$$

This second form tells us that the random interval

$$\left(\bar{X} - 1.96\frac{\sigma}{\sqrt{n}}, \bar{X} + 1.96\frac{\sigma}{\sqrt{n}}\right)$$



will include the unknown parameter with a probability 0.95. Because σ is assumed to be known, both the upper and lower end points can be computed as soon as the sample data is available. Thus, we say that the interval

$$\left(\bar{X} - 1.96\frac{\sigma}{\sqrt{n}}, \bar{X} + 1.96\frac{\sigma}{\sqrt{n}}\right)$$

is a **95% confidence interval for** μ when population variable X is normally distributed and σ known.

We do not need always consider confidence intervals to the choice of a 95% level of confidence. We may wish to specify a different level of probability. We denote this probability by $1-\alpha$ and speak of a $100(1-\alpha)\%$ confidence level. The only change is to replace 1.96 with $z_{\alpha/2}$, where $z_{\alpha/2}$ is a such number that $P(-z_{\alpha/2} < Z < z_{\alpha/2}) = 1-\alpha$ when $Z \sim N(0,1)$.

KEY FACT 8.1. *When population variable X is normally distributed and σ is known, a $100(1-\alpha)\%$ confidence interval for μ is given by*

$$\left(\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right).$$



8.2.2 Large sample confidence interval for μ

We consider now more realistic situation for which the population standard deviation σ is unknown. We require the sample size n to be large, and hence the **central limit theorem** tells us that probability statement

$$P(\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}) = 1 - \alpha.$$

approximately holds, whatever is the underlying population distribution. Also, because n is large, replacing $\frac{\sigma}{\sqrt{n}}$ with is estimator $\frac{s}{\sqrt{n}}$ does not appreciably affect the above probability statement. Hence we have the following Key Fact.

KEY FACT 8.2. *When n is large and σ is unknown, a $100(1-\alpha)\%$ confidence interval for μ is given by*

$$\left(\bar{X} - z_{\alpha/2}\frac{s}{\sqrt{n}}, \bar{X} + z_{\alpha/2}\frac{s}{\sqrt{n}}\right),$$

where s is the sample standard deviation.



Hypothesis Testing

- Basic questions when given data
  - Difference vs. Equivalency
- We have to make decisions even when you are unsure.
- Statistics provides an approach to decision making under uncertainty. Sort of decision making by choosing the same way you would bet. *Maximize expected utility* (subjective value).
- *Hypothesis testing or significance testing is a method for testing a claim or hypothesis about a parameter in a population, using data measured in a sample. In this method, we test some hypothesis by determining the likelihood that a sample statistic could have been selected, if the hypothesis regarding the population parameter were true.*



- In inferential statistics, the **null hypothesis** is a general statement or default position that there is nothing new happening, like there is no association among groups, or no relationship between two measured phenomena.
- Main idea
  - It is difficult to prove that a fact is "right".
  - But it is easy to prove that it is "wrong".
  - The reason is that you only have to find one counter example.



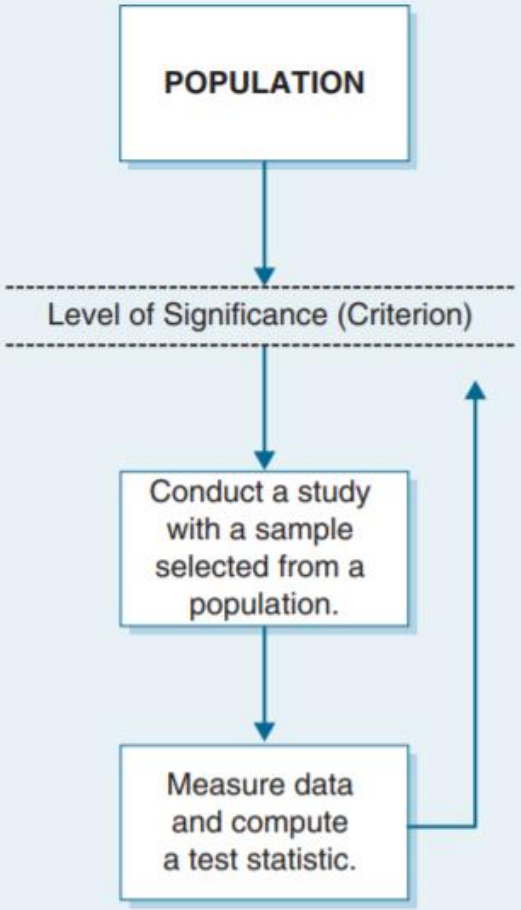
Hypothesis Testing Procedure

STEP 1: State the hypotheses. A researcher states a null hypothesis about a value in the population (H_0) and an alternative hypothesis that contradicts the null hypothesis.

STEP 2: Set the criteria for a decision. A criterion is set upon which a researcher will decide whether to retain or reject the value stated in the null hypothesis.

A sample is selected from the population, and a sample mean is measured.

STEP 3: Compute the test statistic. This will produce a value that can be compared to the criterion that was set before the sample was selected.



STEP 4: Make a decision.
If the probability of obtaining a sample mean is less than 5% when the null is true, then reject the null hypothesis.
If the probability of obtaining a sample mean is greater than 5% when the null is true, then retain the null hypothesis.

Two types of hypothesis

Alternative hypothesis H_1	Null hypothesis H_0
H_1, H_A	H_0
Hypothesis as main purpose of the study	Hypothesis as opposed to purpose
Hypothesis that you want to claim to be right	Hypothesis that you want to claim to be wrong

- It is a way to induce contradiction after assuming null hypothesis is correct.



Null and Alternative Hypotheses

- Idea: It is difficult to prove that a fact is "right", but it is easy to prove that it is "wrong".

- The reason is that you only have to find one counter example.
  - H_1 : All students in DS classes are statistical experts.

- Let's set up the opposite hypothesis
  - H_0 : All students in the DS class do not know statistics.



Null and Alternative Hypotheses

H_1: All students in Convergence Informatics classes are statistical experts.

Our goal is to prove that this proposition is “true”

H_0: All students in the Convergence Informatics class do not know statistics.

If we find evidence for "false" in this null hypothesis, the original proposition is "true."



Null and Alternative Hypotheses

H_1: The height of men and women is different

Our goal is to prove that this proposition is "true"

H_0: The height of men and women is same

- Assuming that men and women are the same height, let's examine the men's and women's heights, respectively, to find the mean and variance.
    - If we find numerical evidence that men and women's height are different in collected data?
    - Original proposition is "true"



Stating a Hypothesis

Example:
Write the claim as a mathematical sentence. State the null and alternative hypotheses and identify which represents the claim.

A manufacturer claims that its rechargeable batteries have an average life of at least 1,000 charges.

Stating a Hypothesis

Example:

Write the claim as a mathematical sentence. State the null and alternative hypotheses and identify which represents the claim.

A manufacturer claims that its rechargeable batteries have an average life of at least 1,000 charges.

$\mu \geq 1000$

Condition of equality

H_0: $\mu \geq 1000$ (Claim)

H_a: $\mu < 1000$

Complement of the null hypothesis

Example:

Write the claim as a mathematical sentence. State the null and alternative hypotheses and identify which represents the claim.

Statesville college claims that 94% of their graduates find employment within six months of graduation.

$p = 0.94$

Condition of equality

H_0: $p = 0.94$ (Claim)

H_a: $p \neq 0.94$

Complement of the null hypothesis

Types of Errors

No matter which hypothesis represents the claim, always begin the hypothesis test **assuming that the null hypothesis is true**.

At the end of the test, one of two decisions will be made:

1. reject the null hypothesis, or
2. fail to reject the null hypothesis.

A **type I error** occurs if the null hypothesis is rejected when it is true.

A **type II error** occurs if the null hypothesis is not rejected when it is false.

Types of Errors

	Actual Truth of H_0	
Decision	**H_0 is true**	**H_0 is false**
Do not reject H_0	Correct Decision	**Type II Error**
Reject H_0	**Type I Error**	Correct Decision

Two types of errors

Example:
Statesville college claims that 94% of their graduates find employment within six months of graduation. What will a type I or type II error be?

H_0: $p = 0.94$ (Claim)

H_a: $p \neq 0.94$

A type I error is rejecting the null when it is true.
The population proportion is actually 0.94, but is rejected. (We believe it is not 0.94.)

A type II error is failing to reject the null when it is false.
The population proportion is not 0.94, but is not rejected. (We believe it is 0.94.)

Two types of errors

- Statistical test results reject null hypothesis or not.

	H_1: signal present	H_0: signal absent
Detection	True Positive	False Positive ***type I error***
Null result	False Negative ***type II error***	True Negative
	TPF = sensitivity = power = TP/(TP+FN)	**FPF = 1-specificity = 1-CL = FP/(FP+TN)**

Confusion Matrix

- Statistical test results reject null hypothesis or not.

"Confusion Matrix" or Contingency Table for Binary Hypothesis Testing

Truth / Declaration	True Hypothesis H_0 (Null)	True Hypothesis H_1
Declared Hypothesis H_0 (Null)	$P(H_0 \mid H_0) = P_{Spec} = Specificity$ $= \frac{\#\, H_0 \text{ Samples Declared } H_0}{\#\, H_0 \text{ Samples}}$	$P(H_0 \mid H_1) = P_{Miss} = P(Miss)$ $= \frac{\#\, H_1 \text{ Samples Declared } H_0}{\#\, H_1 \text{ Samples}}$
Declared Hypothesis H_1	$P(H_1 \mid H_0) = P_{FA} = P(False\ Alarm)$ $= \frac{\#\, H_0 \text{ Samples Declared } H_1}{\#\, H_0 \text{ Samples}}$	$P(H_1 \mid H_1) = P_D = P(Detection)$ $= \frac{\#\, H_1 \text{ Samples Declared } H_1}{\#\, H_1 \text{ Samples}}$
	$P(H_0 \mid H_0) + P(H_1 \mid H_0) = 1$	$P(H_0 \mid H_1) + P(H_1 \mid H_1) = 1$

$$P_{cc} = P(Correct\ Classification) = P(H_0 \mid H_0)P(H_0) + P(H_1 \mid H_1)P(H_1)$$

$$P_e = P(Error) = 1 - P_{cc} = P(H_0 \mid H_1)P(H_1) + P(H_1 \mid H_0)P(H_0)$$

$$\text{Assume: Correct classification is given zero cost} \Rightarrow C_{00} = C_{11} = 0$$

$$\text{Incorrect classification is given full cost} \Rightarrow C_{01} = C_{10} = 1$$

Confusion Matrix

- TPR (True Positive Rate, Power, Sensitivity, Hit rate, Recall)

$$= \frac{TP}{TP+FN} = 1 - \text{FPR}$$

- TNR (True Negative Rate, Specificity)

$$= \frac{TN}{TN+FP} = 1 - \text{FPR}$$

- Be sure to understand and memorize this concept because it is the most widely used measure in statistics & any data science field.

Confusion Matrix

- PPV (Positive Predictive Value, Precision)

$$= \frac{TP}{TP+FP} = 1 - \text{FDR}$$

- NPV (Negative Predictive Value)

$$= \frac{TN}{TN+FN} = 1 - \text{FOR}$$

- FNR (False Negative Rate, Miss rate)

$$= \frac{FN}{FN+TP} = 1 - \text{TPR}$$

- FPR (False Positive Rate, Fall-out)

$$= \frac{FP}{FP+TN} = 1 - \text{TNR}$$

Confusion Matrix

- FDR (False Discovery Rate)

$$= \frac{FP}{FP+TP} = 1 - PPV$$

- ACC (Accuracy)

$$= \frac{TP+TN}{TP+TN+FP+FN}$$

- ERR (Error Rate)

$$= \frac{FP+FN}{TP+TN+FP+FN} = 1 - ACC$$

Level of Significance

In a hypothesis test, the **level of significance** is your maximum allowable probability of making a type I error. It is denoted by α, the lowercase Greek letter alpha.

Hypothesis tests are based on α.

The probability of making a type II error is denoted by β, the lowercase Greek letter beta.

By setting the level of significance at a small value, you are saying that you want the probability of rejecting a true null hypothesis to be small.

Commonly used levels of significance:

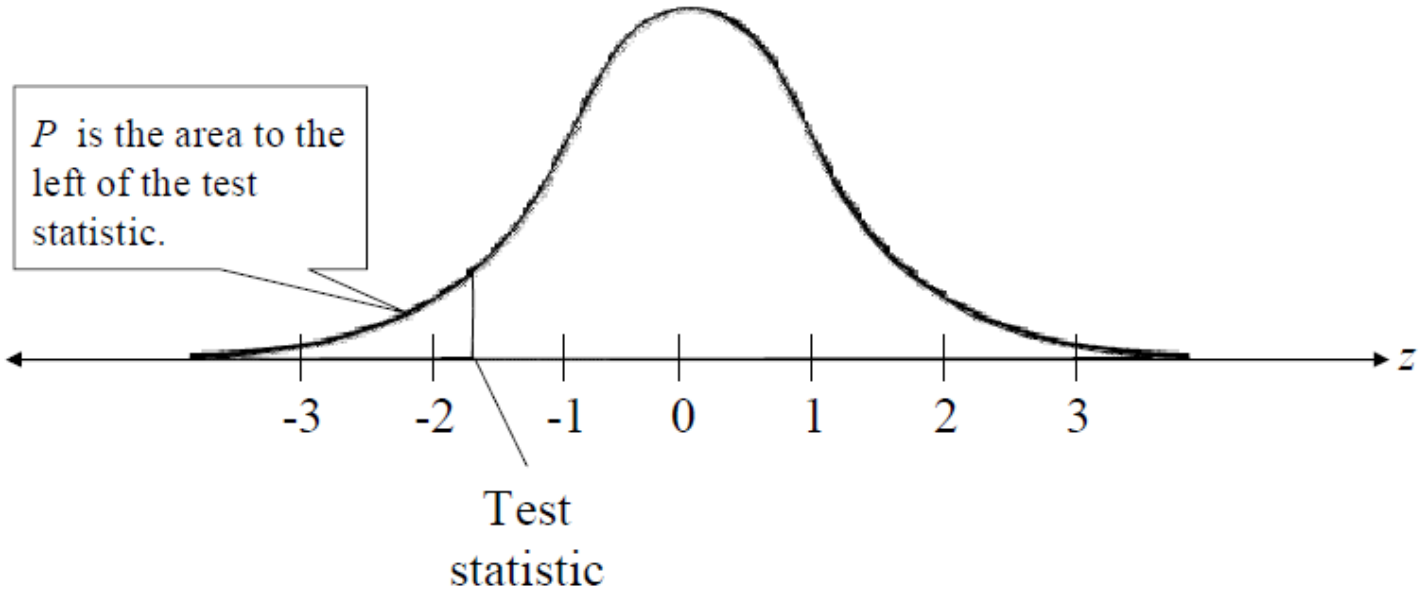
$\alpha = 0.10$ $\alpha = 0.05$ $\alpha = 0.01$

Types of Test: Left-tailed Test

1. If the alternative hypothesis contains the less-than inequality symbol (<), the hypothesis test is a **left-tailed test**.

$H_0: \mu \geq k$

$H_a: \mu < k$

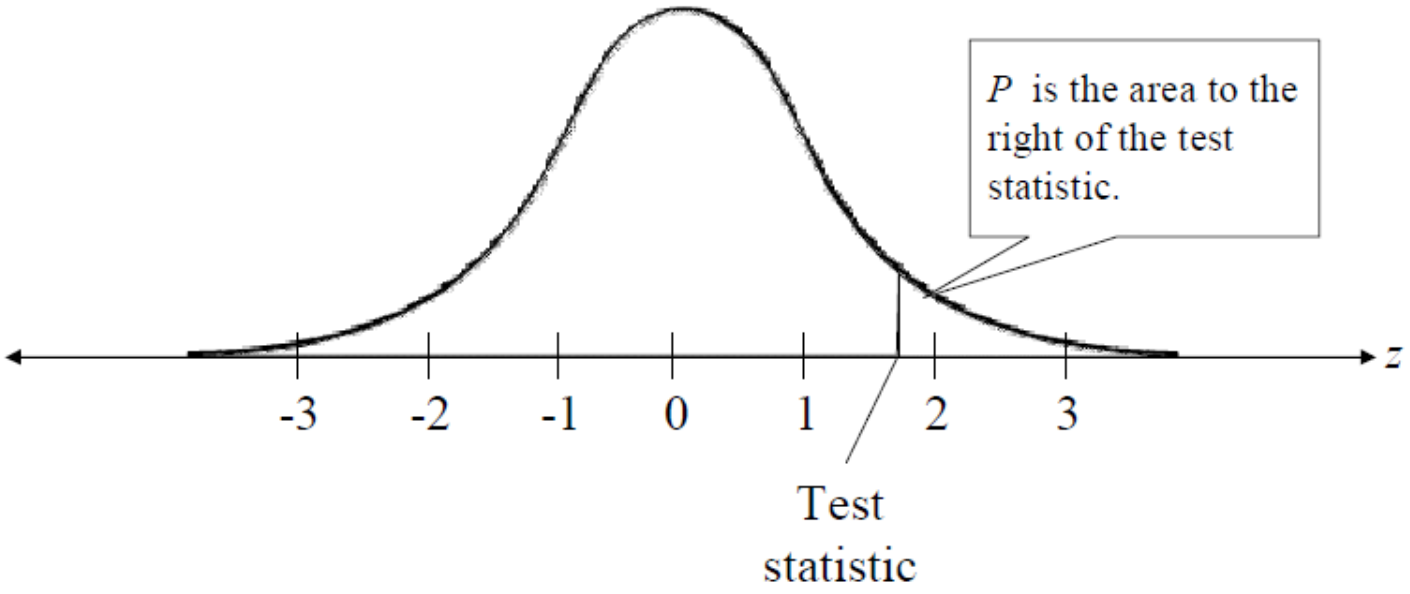


Right-tailed Test

2. If the alternative hypothesis contains the greater-than symbol (>), the hypothesis test is a **right-tailed test**.

$H_0: \mu \leq k$

$H_a: \mu > k$

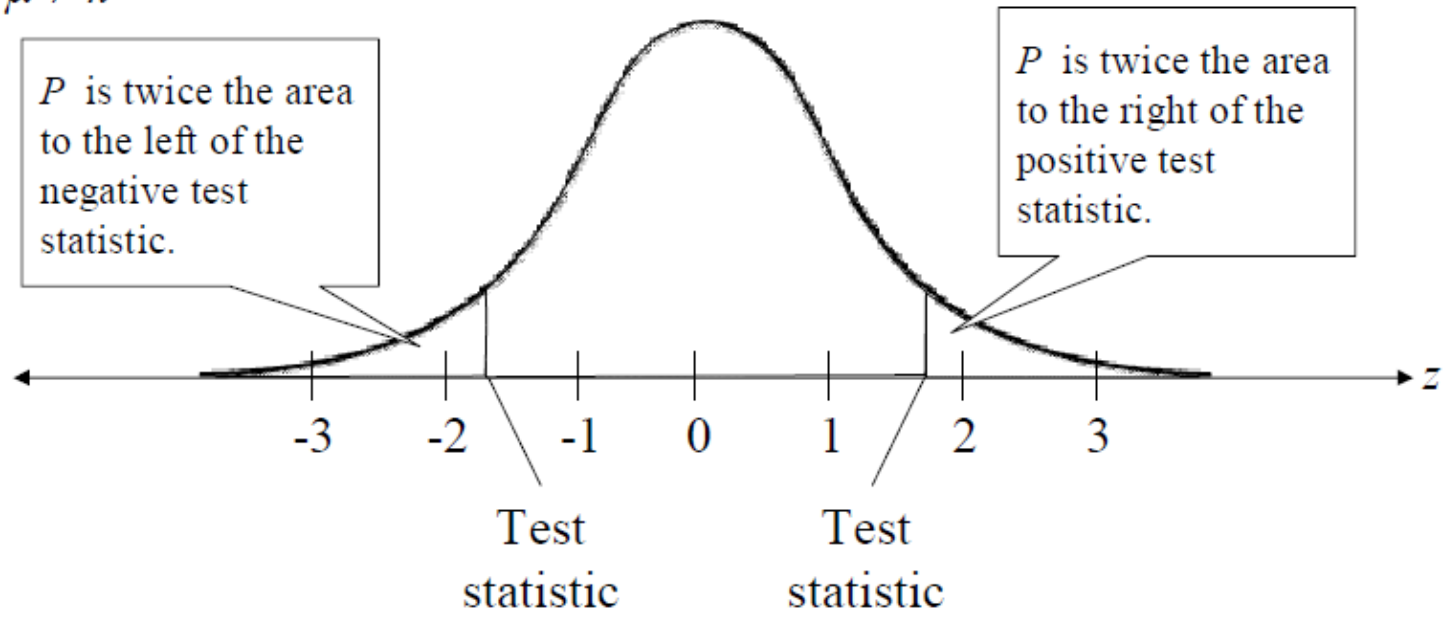


Two-tailed Test

3. If the alternative hypothesis contains the not-equal-to symbol ($\neq$), the hypothesis test is a **two-tailed test**. In a two-tailed test, each tail has an area of $\frac{1}{2}P$.

$H_0: \mu = k$

$H_a: \mu \neq k$



Identifying Types of Tests

Example:

For each claim, state H_0 and H_a. Then determine whether the hypothesis test is a left-tailed, right-tailed, or two-tailed test.

a.) A cigarette manufacturer claims that less than one-eighth of the US adult population smokes cigarettes.

$H_0: p \geq 0.125$

$H_a: p < 0.125$ (Claim) → Left-tailed test

b.) A local telephone company claims that the average length of a phone call is 8 minutes.

$H_0: \mu = 8$ (Claim)

$H_a: \mu \neq 8$ → Two-tailed test

Making a Decision

Decision Rule Based on *P*-value

To use a *P*-value to make a conclusion in a hypothesis test, compare the *P*-value with α.

1. If $P \leq \alpha$, then reject H_0.
2. If $P > \alpha$, then fail to reject H_0.

	Claim	
Decision	**Claim is H_0**	**Claim is H_a**
Reject H_0	There is enough evidence to reject the claim.	There is enough evidence to support the claim.
Do not reject H_0	There is not enough evidence to reject the claim.	There is not enough evidence to support the claim.

Interpreting a Decision

Example:

You perform a hypothesis test for the following claim. How should you interpret your decision if you reject H_0? If you fail to reject H_0?

H_0: (Claim) A cigarette manufacturer claims that less than one-eighth of the US adult population smokes cigarettes.

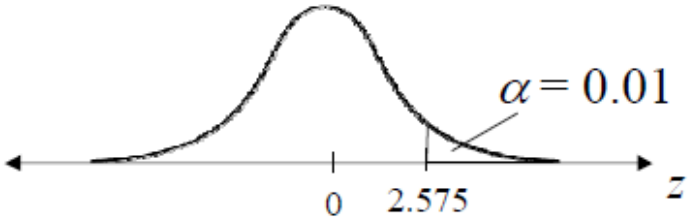
If H_0 is rejected, you should conclude "there is sufficient evidence to indicate that the manufacturer's claim is false."

If you fail to reject H_0, you should conclude "there is *not* sufficient evidence to indicate that the manufacturer's claim is false."

Rejection Regions and Critical Values

A **rejection region** (or **critical region**) of the sampling distribution is the range of values for which the null hypothesis is not probable. If a test statistic falls in this region, the null hypothesis is rejected. A critical value z_0 separates the rejection region from the nonrejection region.

Example:
Find the critical value and rejection region for a right tailed test with $\alpha = 0.01$.



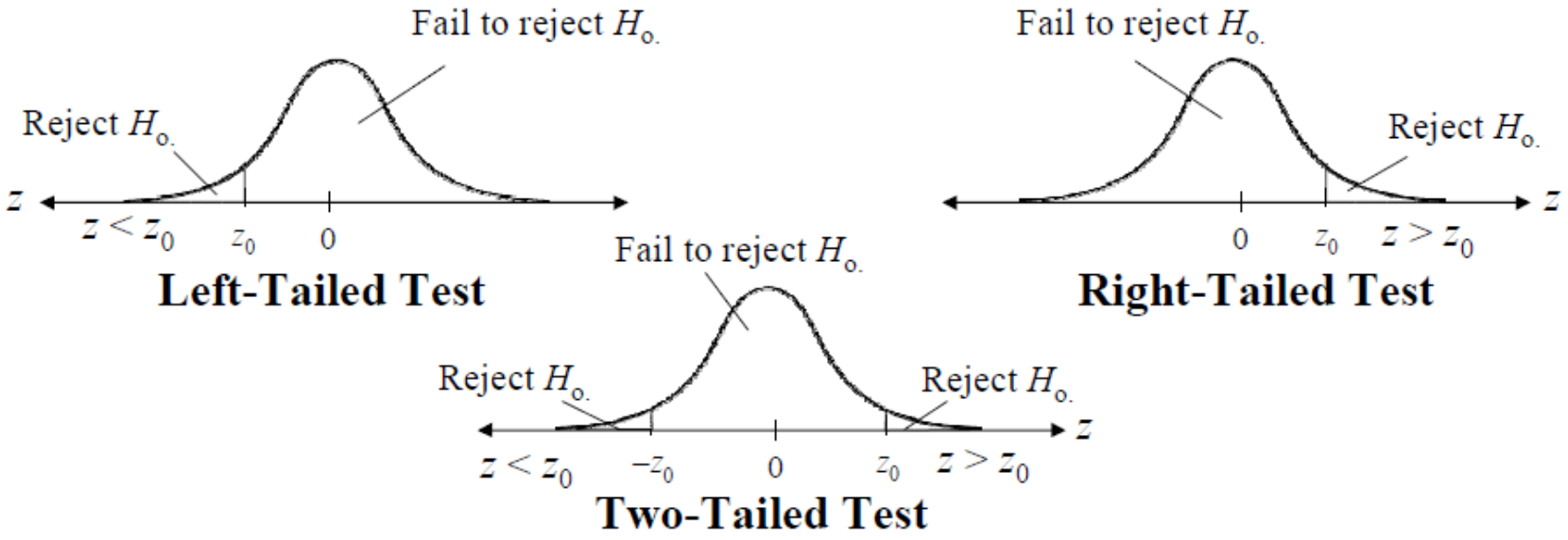
The rejection region is to the right of $z_0 = 2.575$.

Rejection Regions for a z-Test

Decision Rule Based on Rejection Region

To use a rejection region to conduct a hypothesis test, calculate the standardized test statistic, z. If the standardized test statistic

1. is in the rejection region, then reject H_0.
2. is *not* in the rejection region, then fail to reject H_0.



Hypothesis Test Procedure (Traditional Method)

- Step 1 State the hypotheses and identify the claim.
- Step 2 Find the critical value(s) from the appropriate table.
- Step 3 Compute the test value.
- Step 4 Make the decision to reject or not reject the null hypothesis.
- Step 5 Summarize the results.

- Example:

  It has been reported that the average credit card debt for college seniors is $3262. The student senate at a large university feels that their seniors have a debt much less than this, so it conducts a study of 50 randomly selected seniors and finds that the average debt is $2995, and the population standard deviation is $1100. Let's conduct the test based on a Type I error of $\alpha=0.05$.

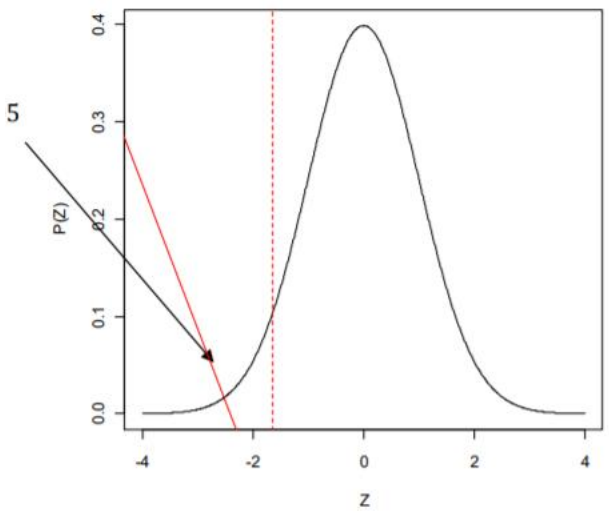
- Step 1 State the hypotheses and identify the claim.

  H_0: $\mu=\$3262$

  H_1: $\mu<\$3262$ **(this is the claim)**

Step 2 Find the critical value(s) from the appropriate table.

Left-tailed test, $\alpha=0.05 \Rightarrow$ Z will be negative and have probability 0.05 underneath it



Z = -1.645 or
Z = -1.65

Table E The Standard Normal Distribution

Cumulative Standard Normal Distribution

z	.00	.01	.02	.03	.04	.05
−3.4	.0003	.0003	.0003	.0003	.0003	.0003
−3.3	.0005	.0005	.0005	.0004	.0004	.0004
−3.2	.0007	.0007	.0006	.0006	.0006	.0006
−3.1	.0010	.0009	.0009	.0009	.0008	.0008
−3.0	.0013	.0013	.0013	.0012	.0012	.0011
−2.9	.0019	.0018	.0018	.0017	.0016	.0016
−2.8	.0026	.0025	.0024	.0023	.0023	.0022
−2.7	.0035	.0034	.0033	.0032	.0031	.0030
−2.6	.0047	.0045	.0044	.0043	.0041	.0040
−2.5	.0062	.0060	.0059	.0057	.0055	.0054
−2.4	.0082	.0080	.0078	.0075	.0073	.0071
−2.3	.0107	.0104	.0102	.0099	.0096	.0094
−2.2	.0139	.0136	.0132	.0129	.0125	.0122
−2.1	.0179	.0174	.0170	.0166	.0162	.0158
−2.0	.0228	.0222	.0217	.0212	.0207	.0202
−1.9	.0287	.0281	.0274	.0268	.0262	.0256
−1.8	.0359	.0351	.0344	.0336	.0329	.0322
−1.7	.0446	.0436	.0427	.0418	.0409	.0401
−1.6	.0548	.0537	.0526	.0516	.0505	.0495

Step 3 Compute the test value.

$$z = \frac{\overline{X} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{2995 - 3262}{\frac{1100}{\sqrt{50}}} = -1.716341$$

Step 4 Make the decision to reject or not reject the null hypothesis.

Since this is a left-tailed test, our rejection region consists of values of Z that are smaller than our critical value of Z = -1.645.

Since our test value (-1.716341) is less than our critical value (-1.645), we reject the null hypothesis.

Step 5 Summarize the results.

We have evidence to support the student senate claim that the university's seniors have credit card debt that is less than the reported average debt.

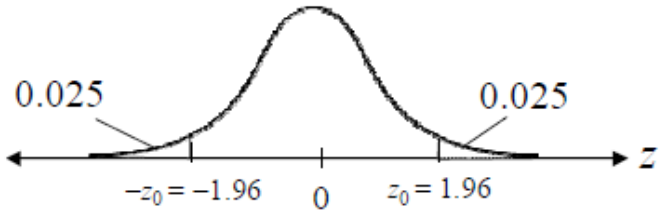
This is based on a Type I error rate of 0.05. This means we falsely make the claim above 5% of the time.

Testing with Rejection Regions

Example:
A local telephone company claims that the average length of a phone call is 8 minutes. In a random sample of 58 phone calls, the sample mean was 7.8 minutes and the standard deviation was 0.5 minutes. Is there enough evidence to support this claim at $\alpha = 0.05$?

H_0: $\mu = 8$ (Claim) $\quad\quad$ H_a: $\mu \neq 8$

The level of significance is $\alpha = 0.05$.



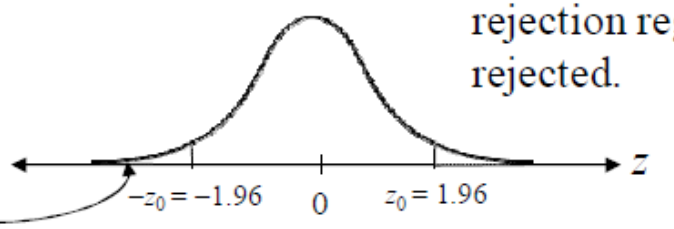
Example continued:

A local telephone company claims that the average length of a phone call is 8 minutes. In a random sample of 58 phone calls, the sample mean was 7.8 minutes and the standard deviation was 0.5 minutes. Is there enough evidence to support this claim at $\alpha = 0.05$?

H_0: $\mu = 8$ (Claim) $\quad$ H_a: $\mu \neq 8$

The standardized test statistic is

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{7.8 - 8}{0.5/\sqrt{58}} \approx -3.05.$$



The test statistic falls in the rejection region, so H_0 is rejected.

At the 5% level of significance, there is enough evidence to reject the claim that the average length of a phone call is 8 minutes.

Null and Alternative Hypotheses (Summery)

- Convert the research question to null and alternative hypotheses
- The **null hypothesis (H_0)** is a claim of "no difference in the population"
- The **alternative hypothesis (H_a)** claims "H_0 is false"
- Collect data and seek evidence against H_0 as a way of bolstering H_a (deduction)



P-Value Method for Hypothesis Testing

- The P-Value (or probability value) is the probability of getting a sample statistic (such as the mean) or a more extreme sample statistic in the direction of the alternative hypothesis when the null hypothesis is true.

A General Rule for Finding P-values using the *Z* distribution or the *t* distribution:

Suppose that z^* is the test statistic of a z test and t^* is the test statistic of a t test

Left-tailed test: p-value $= P(Z \le z^*)$ or p-value $= P(T \le t^*)$
(depending on whether we are doing a z test or a t test).

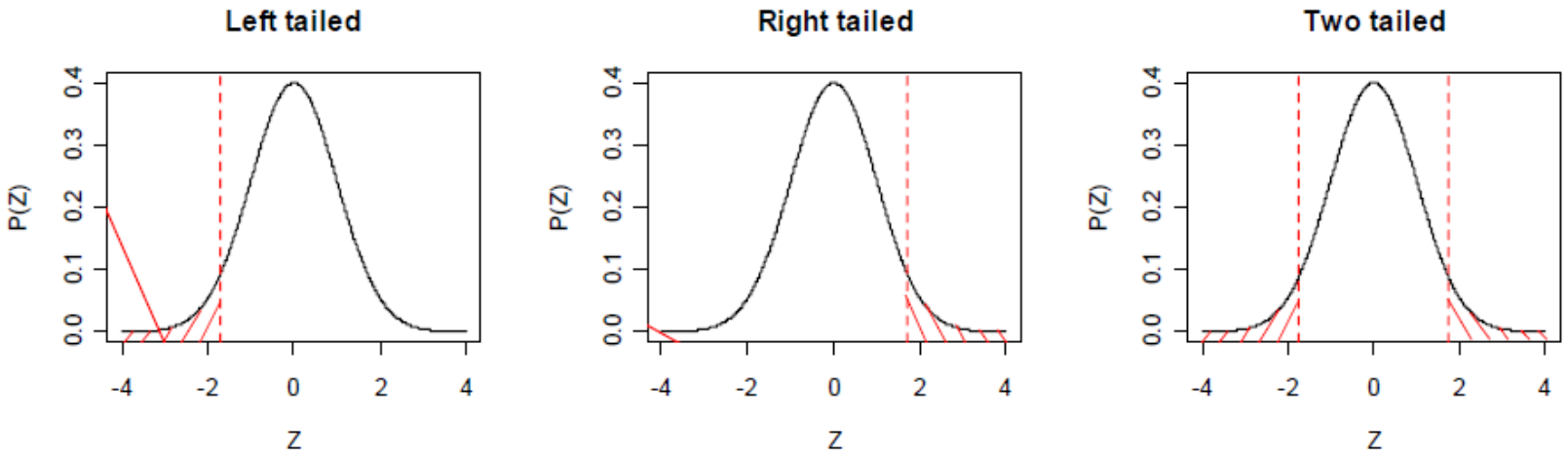
Right-tailed test: p-value $= P(Z \ge z^*)$ or p-value $= P(T \ge t^*)$
(depending on whether we are doing a z test or a t test).

Two-tailed test: p-value $= 2P(Z \ge |z^*|)$ or p-value $= 2P(T \ge |t^*|)$
(depending on whether we are doing a z test or a t test).



The smaller the P-value, the stronger the evidence is against H_0.

We use Table E to find P-Values that use a z Test.



Illustrative Example: "Body Weight"

- **The problem:** In the 1970s, 20–29 year old men in the U.S. had a mean μ body weight of 170 pounds. Standard deviation σ was 40 pounds. We test whether mean body weight in the population now differs.
- **Null hypothesis** $H_{0:}\ \mu = 170$ ("no difference")
- The **alternative hypothesis** can be either

  $H_{a:}\ \mu > 170$ (**one-sided test**) or
  $H_{a:}\ \mu \neq 170$ (**two-sided test**)



Test Statistic

This is an example of a one-sample test of a mean when σ is known. Use this statistic to test the problem:

$$z_{stat} = \frac{\bar{x} - \mu_0}{\sigma_{\bar{x}}}$$

where $\mu_0 \equiv$ population mean assuming H_0 is true

$$\text{and } \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$



Illustrative Example: z statistic

- For the illustrative example, $\mu_0 = 170$
- We know $\sigma = 40$
- Take an SRS of $n = 64$. Therefore

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{40}{\sqrt{64}} = 5$$

- If we found a sample mean of 173, then

$$z_{\text{stat}} = \frac{\bar{x} - \mu_0}{\sigma_{\bar{x}}} = \frac{173 - 170}{5} = 0.60$$



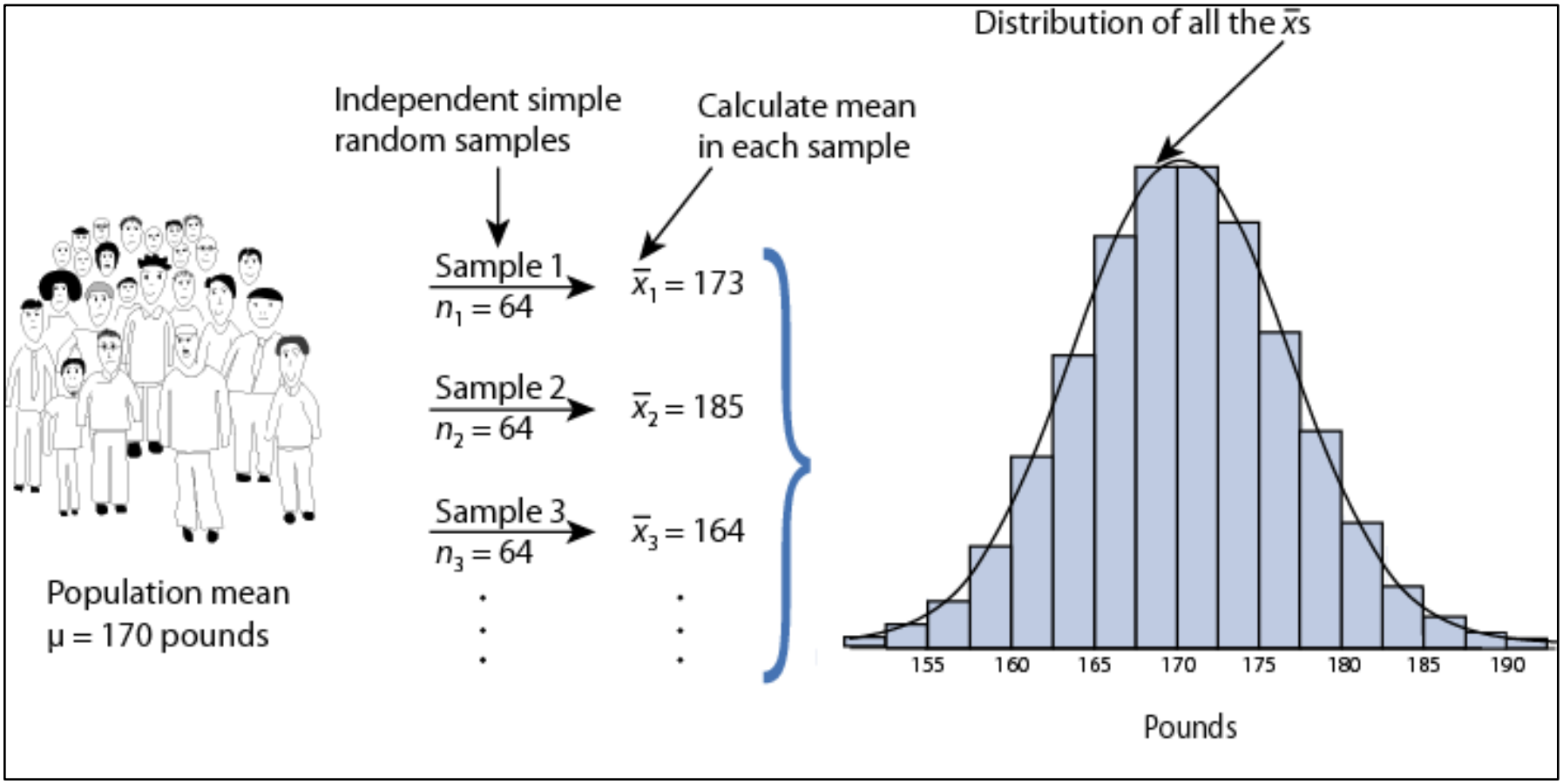
Illustrative Example: z statistic

If we found a sample mean of 185, then

$$z_{\text{stat}} = \frac{\bar{x} - \mu_0}{\sigma_{\bar{x}}} = \frac{185 - 170}{5} = 3.00$$



Reasoning



Sampling distribution of xbar under H_0: μ = 170 for n = 64 $\Rightarrow$ $\bar{x} \sim N(170,5)$

P-value

- The *P*-value answer the question: What is the probability of the observed test statistic or one more extreme **when H_0 *is true*?**
- Convert *z* statistics to *P*-value :

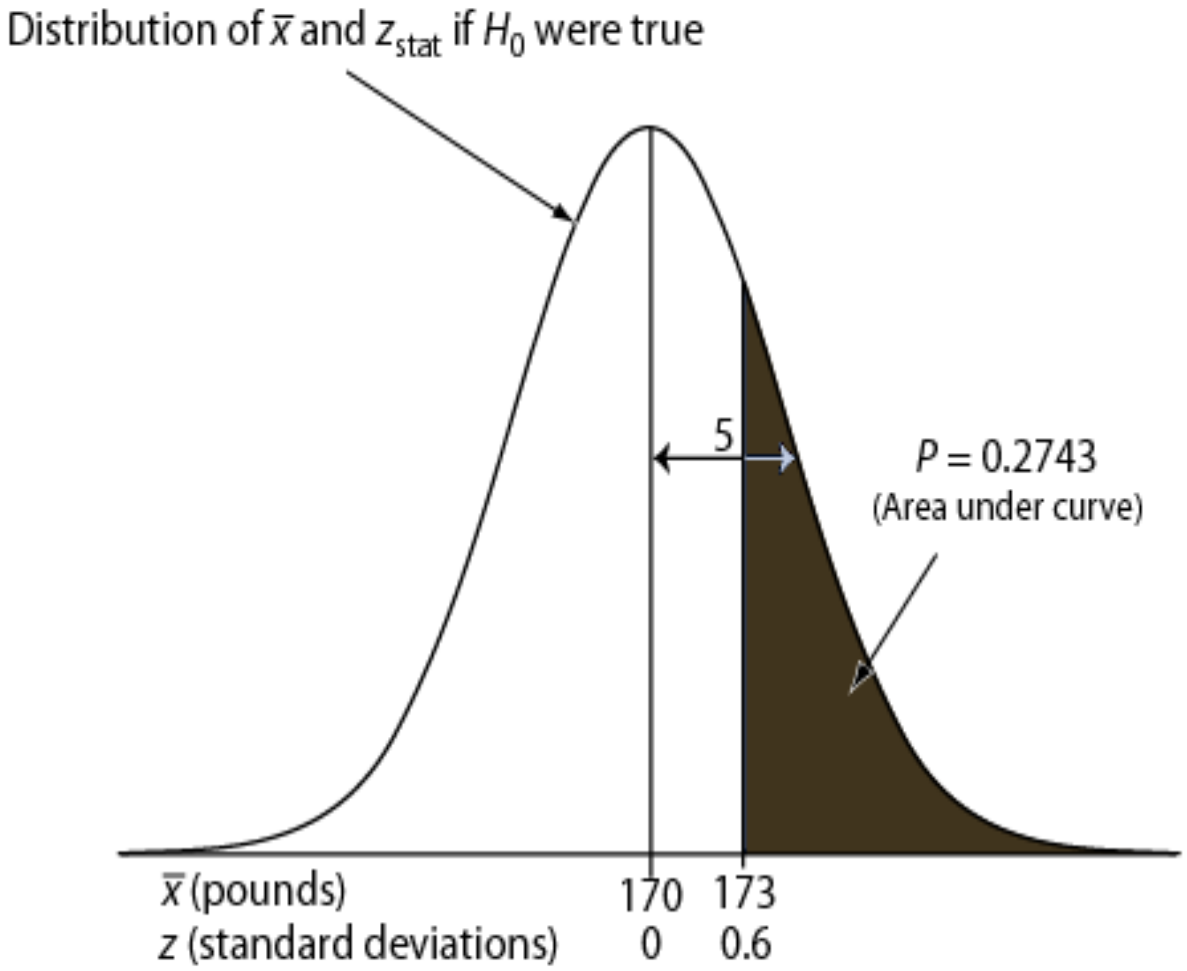
  For $H_a: \mu > \mu_0 \Rightarrow P = \Pr(Z > z_{stat})$ = right-tail beyond z_{stat}

  For $H_a: \mu < \mu_0 \Rightarrow P = \Pr(Z < z_{stat})$ = left tail beyond z_{stat}

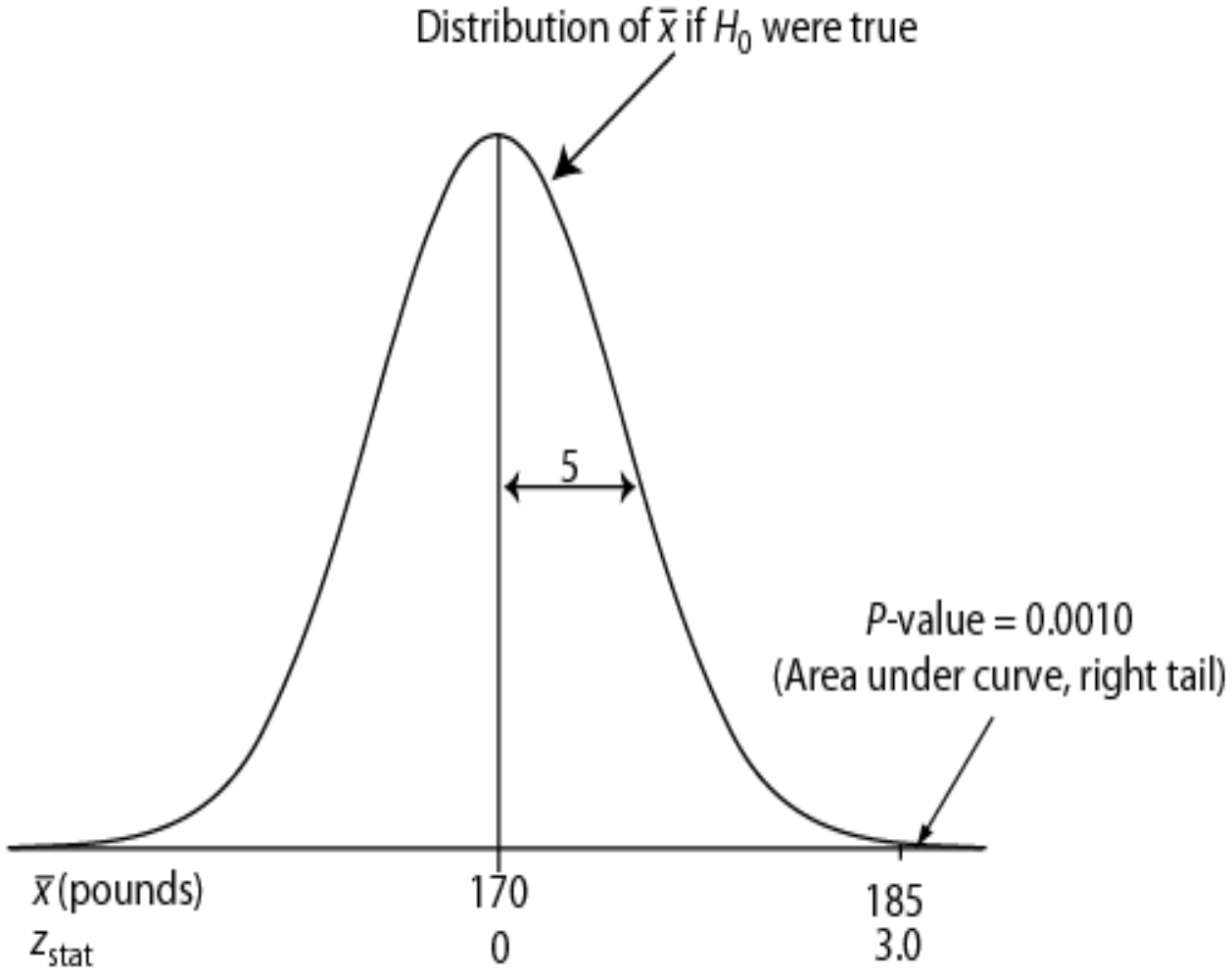
  For $H_a: \mu \neq \mu_0 \Rightarrow P = 2 \times$ one-tailed *P*-value
- Use Table or software to find these probabilities



One-sided *P*-value for z_{stat} of 0.6

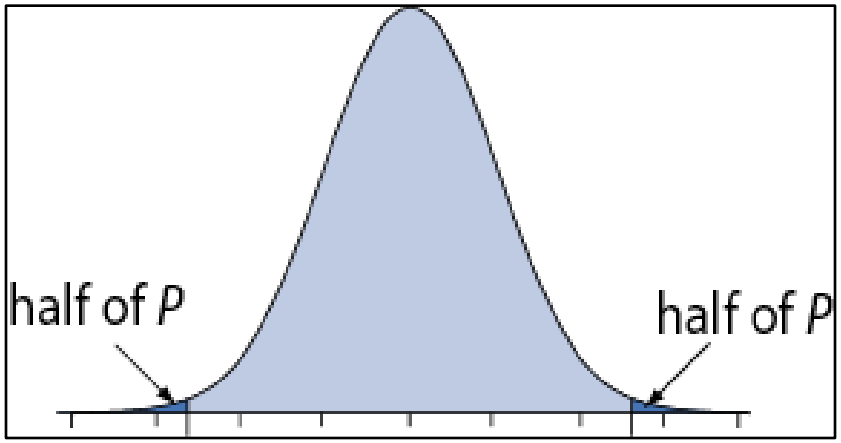


One-sided *P*-value for z_{stat} of 3.0



Two-Sided *P*-Value

- One-sided $H_a \Rightarrow$ AUC in tail beyond z_{stat}
- Two-sided $H_a \Rightarrow$ consider potential deviations in both directions $\Rightarrow$ double the one-sided *P*-value



Examples: If one-sided $P = 0.0010$, then two-sided $P = 2 \times 0.0010 = 0.0020$. If one-sided $P = 0.2743$, then two-sided $P = 2 \times 0.2743 = 0.5486$.



Interpretation

- *P*-value answer the question: What is the probability of the observed test statistic … **when H_0 *is true*?**
- Thus, smaller and smaller *P*-values provide stronger and stronger evidence against H_0
- Small *P*-value $\Rightarrow$ strong evidence



Interpretation

Conventions*

$P > 0.10 \Rightarrow$ non-significant evidence against H_0

$0.05 < P \leq 0.10 \Rightarrow$ marginally significant evidence

$0.01 < P \leq 0.05 \Rightarrow$ significant evidence against H_0

$P \leq 0.01 \Rightarrow$ highly significant evidence against H_0

Examples

$P = .27 \Rightarrow$ non-significant evidence against H_0

$P = .01 \Rightarrow$ highly significant evidence against H_0

*** It is *unwise* to draw firm borders for "significance"**

α-Level (Used in some situations)

- Let α ≡ probability of erroneously rejecting H_0
- Set α threshold (e.g., let α = .10, .05, *or whatever*)
- Reject H_0 when $P \leq \alpha$
- Retain H_0 when $P > \alpha$
- Example: Set α = .10. Find $P = 0.27 \Rightarrow$ retain H_0
- Example: Set α = .01. Find $P = .001 \Rightarrow$ reject H_0



Conditions for z test

- σ known (not from data)
- Population approximately Normal or large sample (central limit theorem)
- Data valid



Steps for Hypothesis Testing

1. State the claim mathematically and verbally. Identify the null and alternative hypotheses.

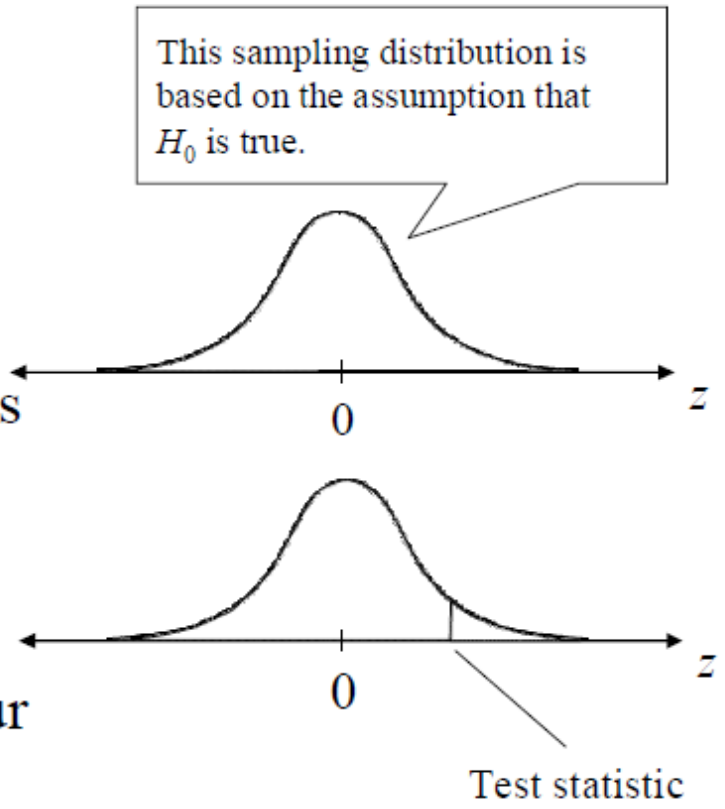
   H_0: ? H_a: ?

2. Specify the level of significance.

   $\alpha = ?$

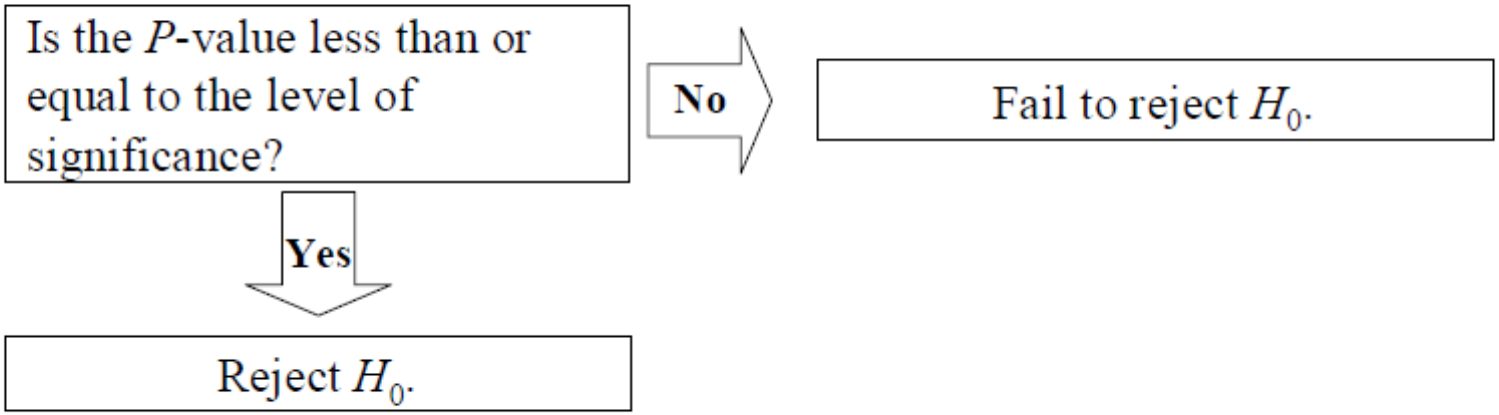
3. Determine the standardized sampling distribution and draw its graph.

4. Calculate the test statistic and its standardized value. Add it to your sketch.



5. Find the *P*-value.

6. Use the following decision rule.



7. Write a statement to interpret the decision in the context of the original claim.

Hypothesis Testing for the Mean (Large Samples)

Decision Rule Based on *P*-value

To use a *P*-value to make a conclusion in a hypothesis test, compare the *P*-value with α.

1. If $P \leq \alpha$, then reject H_0.
2. If $P > \alpha$, then fail to reject H_0.

Recall that when the sample size is at least 30, the sampling distribution for the sample mean is normal.



Using P-values to Make a Decision

Example:

The *P*-value for a hypothesis test is $P = 0.0256$. What is your decision if the level of significance is

a.) 0.05,

b.) 0.01?

a.) Because 0.0256 is < 0.05, you should reject the null hypothesis.

b.) Because 0.0256 is > 0.01, you should fail to reject the null hypothesis.



Finding the P-value

After determining the hypothesis test's standardized test statistic and the test statistic's corresponding area, do one of the following to find the *P*-value.

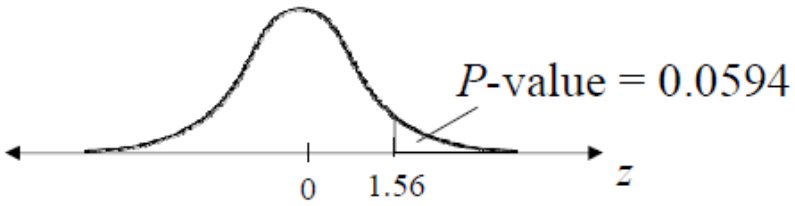
a. For a left-tailed test, P = (Area in left tail).

b. For a right-tailed test, P = (Area in right tail).

c. For a two-tailed test, $P = 2$(Area in tail of test statistic).

Example:

The test statistic for a right-tailed test is $z = 1.56$. Find the *P*-value.

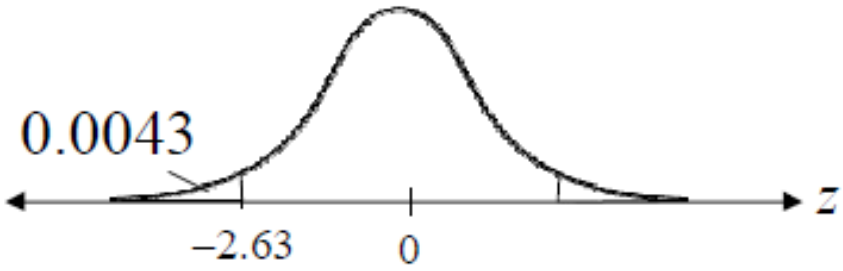


The area to the right of z = 1.56 is 1 – .9406 = 0.0594.

Example:

The test statistic for a two-tailed test is $z = -2.63$.

d the P-value.



The area to the left of $z = -2.63$ is 0.0043.

The P-value is $2(0.0043) = 0.0086$

Using P-values for a z-Test

The ***z*-test for the mean** is a statistical test for a population mean. The *z*-test can be used when the population is normal and σ is known, or for any population when the sample size n is at least 30.

The **test statistic** is the sample mean $\bar{x}$ and the **standardized test statistic** is z.

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} \qquad \frac{\sigma}{\sqrt{n}} = \text{standard error} = \sigma_{\bar{x}}$$

When $n \geq 30$, the sample standard deviation s can be substituted for σ.



Using P-values for a z-Test

Using *P*-values for a *z*-Test for a Mean μ

In Words	*In Symbols*
1. State the claim mathematically and verbally. Identify the null and alternative hypotheses.	State H_0 and H_a.
2. Specify the level of significance.	Identify α.
3. Determine the standardized test statistic.	$z = \dfrac{\bar{x} - \mu}{\sigma / \sqrt{n}}$
4. Find the area that corresponds to z.	Use Table 4 in Appendix B.



Using *P*-values for a *z*-Test for a Mean μ

In Words *In Symbols*

5. Find the *P*-value.
   a. For a left-tailed test, P = (Area in left tail).
   b. For a right-tailed test, P = (Area in right tail).
   c. For a two-tailed test, P = 2(Area in tail of test statistic).

6. Make a decision to reject or fail to reject the null hypothesis.

   Reject H_0 if *P*-value is less than or equal to α. Otherwise, fail to reject H_0.

7. Interpret the decision in the context of the original claim.



Hypothesis Testing with P-values

Example:

A manufacturer claims that its rechargeable batteries are good for an average of more than 1,000 charges. A random sample of 100 batteries has a mean life of 1002 charges and a standard deviation of 14. Is there enough evidence to support this claim at $\alpha = 0.01$?

$H_0: \mu \leq 1000$ $\qquad$ $H_a: \mu > 1000$ (Claim)

The level of significance is $\alpha = 0.01$.

The standardized test statistic is

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{1002 - 1000}{14 / \sqrt{100}}$$

$$\approx 1.43$$

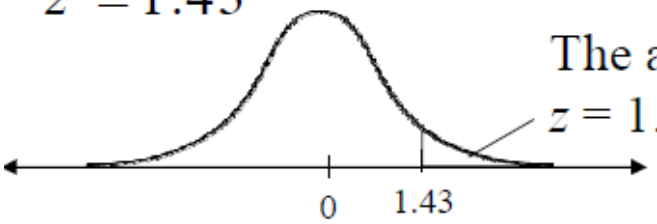


Example continued:

A manufacturer claims that its rechargeable batteries are good for an average of more than 1,000 charges. A random sample of 100 batteries has a mean life of 1002 charges and a standard deviation of 14. Is there enough evidence to support this claim at $\alpha = 0.01$?

$H_0: \mu \leq 1000$ $\quad$ $H_a: \mu > 1000$ (Claim)

$z = 1.43$



The area to the right of $z = 1.43$ is $P = 0.0764$.

P-value is greater than $\alpha = 0.01$, fail to reject H_0.

At the 1% level of significance, there is not enough evidence to support the claim that the rechargeable battery has an average life of at least 1000 charges.

Hypothesis Testing for the Mean (Small Samples)

- Critical Values in a t-Distribution

Finding Critical Values in a *t*-Distribution

1. Identify the level of significance α.
2. Identify the degrees of freedom d.f. = $n - 1$.
3. Find the critical value(s) using Table 5 in Appendix B in the row with $n - 1$ degrees of freedom. If the hypothesis test is
   a. left-tailed, use "One Tail, α" column with a negative sign,
   b. right-tailed, use "One Tail, α" column with a positive sign,
   c. two-tailed, use "Two Tails, α" column with a negative and a positive sign.



Finding Critical Values for t

Example:
Find the critical value t_0 for a right-tailed test given $\alpha = 0.01$ and $n = 24$.

The degrees of freedom are d.f. = $n - 1 = 24 - 1 = 23$.

To find the critical value, use Table 5 with d.f. = 23 and 0.01 in the "One Tail, α" column. Because the test is a right-tail test, the critical value is positive.

$t_0 = 2.500$



Example:
Find the critical values t_0 and $-t_0$ for a two-tailed test given $\alpha = 0.10$ and $n = 12$.

The degrees of freedom are d.f. = $n - 1 = 12 - 1 = 11$.

To find the critical value, use Table 5 with d.f. = 11 and 0.10 in the "Two Tail, α" column. Because the test is a two-tail test, one critical value is negative and one is positive.

$-t_0 = -1.796$ and $t_0 = 1.796$



t-Test for a Mean μ (n < 30, Unknown)

The ***t*-test for the mean** is a statistical test for a population mean. The *t*-test can be used when the population is normal or nearly normal, σ is unknown, and $n < 30$.

The **test statistic** is the sample mean $\bar{x}$ and the **standardized test statistic** is *t*.

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

The degrees of freedom are d.f. = $n - 1$.



Using the *t*-Test for a Mean μ (Small Sample)

In Words	*In Symbols*
1. State the claim mathematically and verbally. Identify the null and alternative hypotheses.	State H_0 and H_a.
2. Specify the level of significance.	Identify α.
3. Identify the degrees of freedom and sketch the sampling distribution.	d.f. $= n - 1$.
4. Determine any critical values.	Use Table 5 in Appendix B.
5. Determine any rejection region(s).	



Using the *t*-Test for a Mean μ (Small Sample)

In Words	*In Symbols*
6. Find the standardized test statistic.	$t = \dfrac{\bar{x} - \mu}{s/\sqrt{n}}$
7. Make a decision to reject or fail to reject the null hypothesis.	If t is in the rejection region, reject H_0. Otherwise, fail to reject H_0.
8. Interpret the decision in the context of the original claim.	



Example:
A local telephone company claims that the average length of a phone call is 8 minutes. In a random sample of 18 phone calls, the sample mean was 7.8 minutes and the standard deviation was 0.5 minutes. Is there enough evidence to support this claim at $\alpha = 0.05$?

$H_0: \mu = 8$ (Claim) $\quad\quad$ $H_a: \mu \neq 8$

The level of significance is $\alpha = 0.05$.

The test is a two-tailed test.

Degrees of freedom are d.f. = 18 – 1 = 17.

The critical values are $-t_0 = -2.110$ and $t_0 = 2.110$



Example continued:

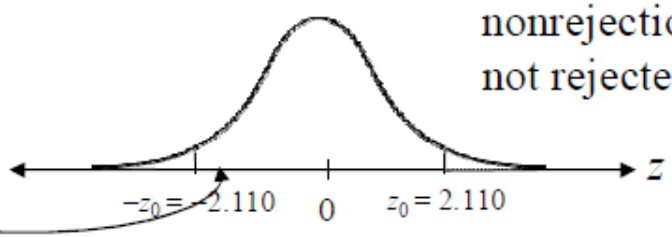
A local telephone company claims that the average length of a phone call is 8 minutes. In a random sample of 18 phone calls, the sample mean was 7.8 minutes and the standard deviation was 0.5 minutes. Is there enough evidence to support this claim at $\alpha = 0.05$?

H_0: $\mu = 8$ (Claim) $\quad\quad$ H_a: $\mu \neq 8$

The standardized test statistic is

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{7.8 - 8}{0.5/\sqrt{18}}$$

$$\approx -1.70.$$



The test statistic falls in the nonrejection region, so H_0 is not rejected.

At the 5% level of significance, there is not enough evidence to reject the claim that the average length of a phone call is 8 minutes.

Testing μ Using P-values

Example:

A manufacturer claims that its rechargeable batteries have an average life greater than 1,000 charges. A random sample of 10 batteries has a mean life of 1002 charges and a standard deviation of 14. Is there enough evidence to support this claim at $\alpha = 0.01$?

$H_0: \mu \leq 1000$ $\qquad$ $H_a: \mu > 1000$ (Claim)

The level of significance is $\alpha = 0.01$.

The degrees of freedom are d.f. $= n - 1 = 10 - 1 = 9$.

The standardized test statistic is

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{1002 - 1000}{14/\sqrt{10}} \approx 0.45$$

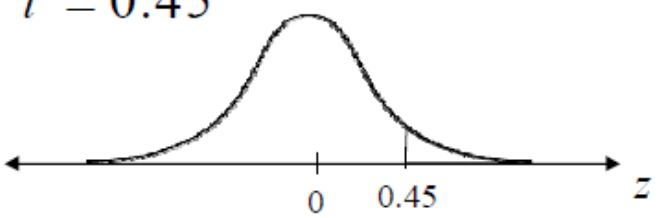
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Example continued:

A manufacturer claims that its rechargeable batteries have an average life greater than 1,000 charges. A random sample of 10 batteries has a mean life of 1002 charges and a standard deviation of 14. Is there enough evidence to support this claim at $\alpha = 0.01$?

$H_0: \mu \leq 1000$ $\quad$ $H_a: \mu > 1000$ (Claim)

$t = 0.45$



Using the d.f. = 9 row from Table 5, you can determine that P is greater than $\alpha = 0.25$ and is therefore also greater than the 0.01 significance level. H_0 would fail to be rejected.

At the 1% level of significance, there is not enough evidence to support the claim that the rechargeable battery has an average life of at least 1000 charges.

Hypothesis Testing for Proportions

The ***z*-test for a population** is a statistical test for a population proportion. The *z*-test can be used when a binomial distribution is given such that $np \geq 5$ and $nq \geq 5$.

The **test statistic** is the sample proportion $\hat{p}$ and the **standardized test statistic** is *z*.

$$z = \frac{\hat{p} - \mu_{\hat{p}}}{\sigma_{\hat{p}}} = \frac{\hat{p} - p}{\sqrt{pq/n}}$$



Hypothesis Test for Proportions

Using a *z*-Test for a Proportion *p*

Verify that $np \geq 5$ and $nq \geq 5$.

	In Words	*In Symbols*
1.	State the claim mathematically and verbally. Identify the null and alternative hypotheses.	State H_0 and H_a.
2.	Specify the level of significance.	Identify α.
3.	Sketch the sampling distribution.	
4.	Determine any critical values.	Use Table 4 in Appendix B.



Using a *z*-Test for a Proportion *p*

Verify that $np \geq 5$ and $nq \geq 5$.

In Words

5. Determine any rejection regions.
6. Find the standardized test statistic.
7. Make a decision to reject or fail to reject the null hypothesis.
8. Interpret the decision in the context of the original claim.

In Symbols

$$z = \frac{\hat{p} - p}{\sqrt{pq/n}}$$

If z is in the rejection region, reject H_0. Otherwise, fail to reject H_0.



Example:
Statesville college claims that more than 94% of their graduates find employment within six months of graduation. In a sample of 500 randomly selected graduates, 475 of them were employed. Is there enough evidence to support the college's claim at a 1% level of significance?

Verify that the products np and nq are at least 5.

$$np = (500)(0.94) = 470 \quad \text{and} \quad nq = (500)(0.06) = 30$$

$H_0: p \leq 0.94$ $\qquad$ $H_a: p > 0.94$ (Claim)



Example continued:

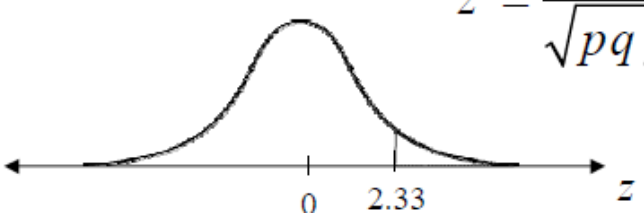
Statesville college claims that more than 94% of their graduates find employment within six months of graduation. In a sample of 500 randomly selected graduates, 475 of them were employed. Is there enough evidence to support the college's claim at a 1% level of significance?

$H_0: p \leq 0.94$ $\qquad$ $H_a: p > 0.94$ (Claim)

Because the test is a right-tailed test and $\alpha = 0.01$, the critical value is 2.33.

$$z = \frac{\hat{p} - p}{\sqrt{pq/n}} = \frac{0.95 - 0.94}{\sqrt{(0.94)(0.06)/500}}$$

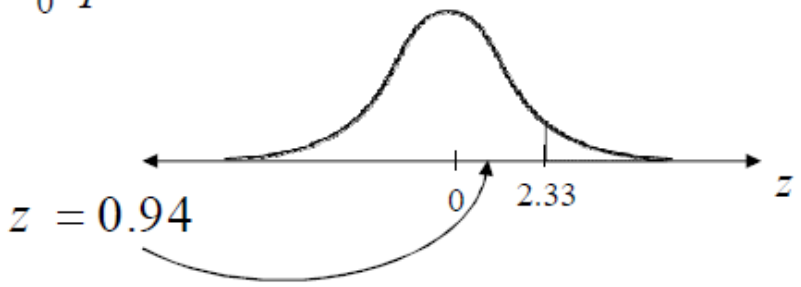
$$\approx 0.94$$ Test statistic



Example continued:

Statesville college claims that more than 94% of their graduates find employment within six months of graduation. In a sample of 500 randomly selected graduates, 475 of them were employed. Is there enough evidence to support the college's claim at a 1% level of significance?

$H_0: p \leq 0.94$ $\quad$ $H_a: p > 0.94$ (Claim)



The test statistic falls in the nonrejection region, so H_0 is not rejected.

At the 1% level of significance, there is not enough evidence to support the college's claim.

Example:
A cigarette manufacturer claims that one-eighth of the US adult population smokes cigarettes. In a random sample of 100 adults, 5 are cigarette smokers. Test the manufacturer's claim at $\alpha = 0.05$.

Verify that the products np and nq are at least 5.

$$np = (100)(0.125) = 12.5 \text{ and } nq = (100)(0.875) = 87.5$$

$H_0: p = 0.125$ (Claim) $\quad$ $H_a: p \neq 0.125$

Because the test is a two-tailed test and $\alpha = 0.05$, the critical values are ± 1.96.

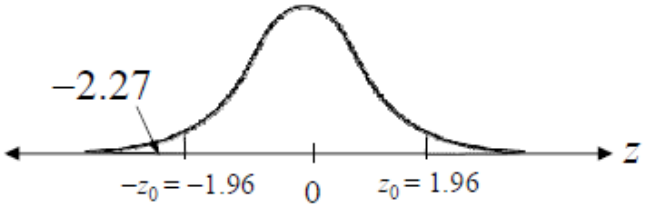


Example continued:

A cigarette manufacturer claims that one-eighth of the US adult population smokes cigarettes. In a random sample of 100 adults, 5 are cigarettes smokers. Test the manufacturer's claim at $\alpha = 0.05$.

H_0: $p = 0.125$ (Claim)

H_a: $p \neq 0.125$



The test statistic is

$$z = \frac{\hat{p} - p}{\sqrt{pq/n}} = \frac{0.05 - 0.125}{\sqrt{(0.125)(0.875)/100}}$$

≈ -2.27 Reject H_0.

At the 5% level of significance, there is enough evidence to reject the claim that one-eighth of the population smokes.

Hypothesis Testing for Variance and Standard Deviation

- When analyzing numerical data, sometimes you need to draw conclusions about the population variance or standard deviation.

The **χ^2-test for a variance or standard deviation** is a statistical test for a population variance or standard deviation. The χ^2-test can be used when the population is normal.

The **test statistic** is s^2 and the **standardized test statistic**

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2}$$

follows a chi-square distribution with degrees of freedom d.f. = $n - 1$.



Using the χ^2-Test for a Variance or Standard Deviation

	In Words	*In Symbols*
1.	State the claim mathematically and verbally. Identify the null and alternative hypotheses.	State H_0 and H_a.
2.	Specify the level of significance.	Identify α.
3.	Determine the degrees of freedom and sketch the sampling distribution.	d.f. $= n - 1$
4.	Determine any critical values.	Use Table 6 in Appendix B.



Using the χ2-Test for a Variance or Standard Deviation

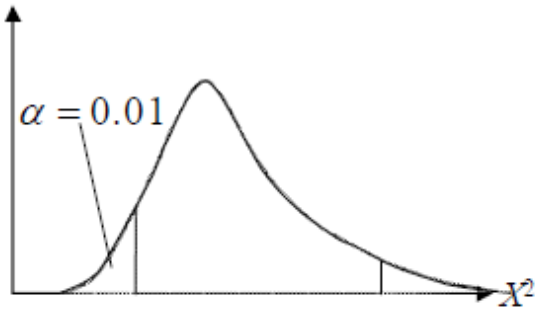
In Words	*In Symbols*
5. Determine any rejection regions.	
6. Find the standardized test statistic.	$\chi^2 = \frac{(n-1)s^2}{\sigma^2}$
7. Make a decision to reject or fail to reject the null hypothesis.	If χ^2 is in the rejection region, reject H_0. Otherwise, fail to reject H_0.
8. Interpret the decision in the context of the original claim.	



A college professor claims that the standard deviation for students taking a statistics test is less than 30. 10 tests are randomly selected and the standard deviation is found to be 28.8 Test this professor's claim at the $\alpha = 0.01$ level.

H_0: $\sigma \geq 30$ $\quad\quad$ H_a: $\sigma < 30$ (Claim)

This is a left-tailed test with d.f.= 9 and $\alpha = 0.01$.



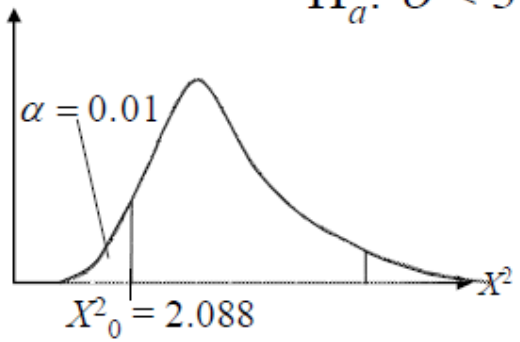
Example continued:

A college professor claims that the standard deviation for students taking a statistics test is less than 30. 10 tests are randomly selected and the standard deviation is found to be 28.8. Test this professor's claim at the $\alpha = 0.01$ level.

H_0: $\sigma \geq 30$

H_a: $\sigma < 30$ (Claim)

$\chi^2_0 = 2.088$



$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} = \frac{(10-1)(28.8)^2}{30^2} \approx 8.29$$

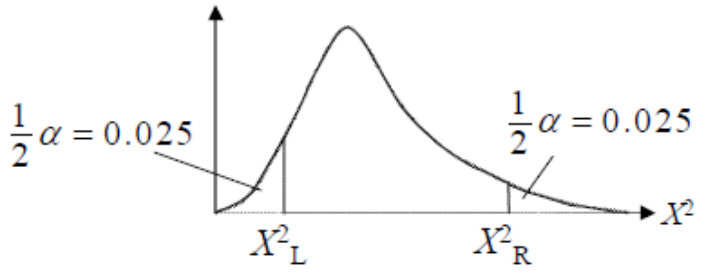
Fail to reject H_0.

At the 1% level of significance, there is not enough evidence to support the professor's claim.

Example:
A local balloon company claims that the variance for the time its helium balloons will stay afloat is 5 hours. A disgruntled customer wants to test this claim. She randomly selects 23 customers and finds that the variance of the sample is 4.5 seconds. At $\alpha = 0.05$, does she have enough evidence to reject the company's claim?

H_0: $\sigma^2 = 5$ (Claim) $\quad\quad$ H_a: $\sigma^2 \neq 5$

This is a two-tailed test with d.f.= 22 and $\alpha = 0.05$.



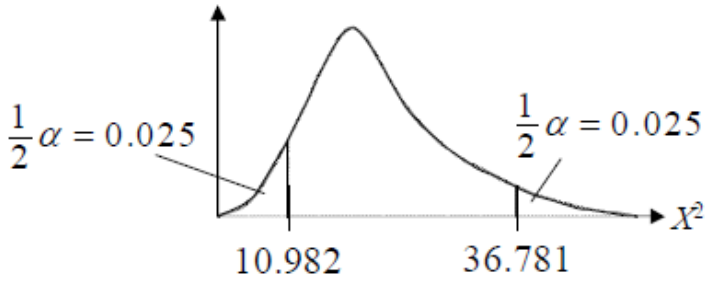
Continued.

Example continued:

A local balloon company claims that the variance for the time its helium balloons will stay afloat is 5 hours. A disgruntled customer wants to test this claim. She randomly selects 23 customers and finds that the variance of the sample is 4.5 seconds. At $\alpha = 0.05$, does she have enough evidence to reject the company's claim?

H_0: $\sigma^2 = 5$ (Claim) $\quad\quad$ H_a: $\sigma^2 \neq 5$

The critical values are $\chi^2_L = 10.982$ and $\chi^2_R = 36.781$.



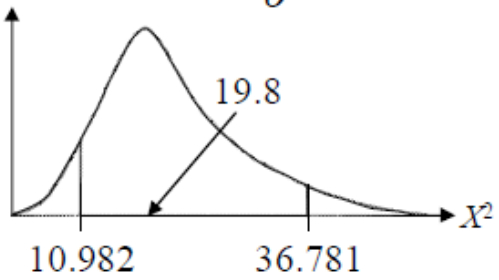
Continued.

Example continued:

A local balloon company claims that the variance for the time one of its helium balloons will stay afloat is 5 hours. A disgruntled customer wants to test this claim. She randomly selects 23 customers and finds that the variance of the sample is 4.5 seconds. At $\alpha = 0.05$, does she have enough evidence to reject the company's claim?

H_0: $\sigma^2 = 5$ (Claim) $\quad$ H_a: $\sigma^2 \neq 5$

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} = \frac{(23-1)(4.5)}{5} = 19.8$$ Fail to reject H_0.



At $\alpha = 0.05$, there is not enough evidence to reject the claim that the variance of the float time is 5 hours.

Two Sample Hypothesis Testing
Testing the Difference Between Means (Large Independent Samples)

- In a two-sample hypothesis test, two parameters from two populations are compared.
- For a two-sample hypothesis test,
  1. the **null hypothesis H_0** is a statistical hypothesis that usually states there is no difference between the parameters of two populations. The null hypothesis always contains the symbol $\leq$, $=$, or $\geq$.
  2. the **alternative hypothesis H_a** is a statistical hypothesis that is true when H_0 is false. The alternative hypothesis always contains the symbol $>$, $\neq$, or $<$.



Two Sample Hypothesis Testing

- To write a null and alternative hypothesis for a two-sample hypothesis test, translate the claim made about the population parameters from a verbal statement to a mathematical statement.

$$\begin{cases} H_0: \mu_1 = \mu_2 \\ H_a: \mu_1 \neq \mu_2 \end{cases}$$

$$\begin{cases} H_0: \mu_1 \leq \mu_2 \\ H_a: \mu_1 > \mu_2 \end{cases}$$

$$\begin{cases} H_0: \mu_1 \geq \mu_2 \\ H_a: \mu_1 < \mu_2 \end{cases}$$

Regardless of which hypotheses used, $\mu_1 = \mu_2$ is always assumed to be true.



Two Sample *z*-Test

Three conditions are necessary to perform a *z*-test for the difference between two population means μ_1 and μ_2.

1. The samples must be randomly selected.
2. The samples must be independent. Two samples are **independent** if the sample selected from one population is not related to the sample selected from the second population.
3. Each sample size must be at least 30, or, if not, each population must have a normal distribution with a known standard deviation.



Two Sample *z*-Test

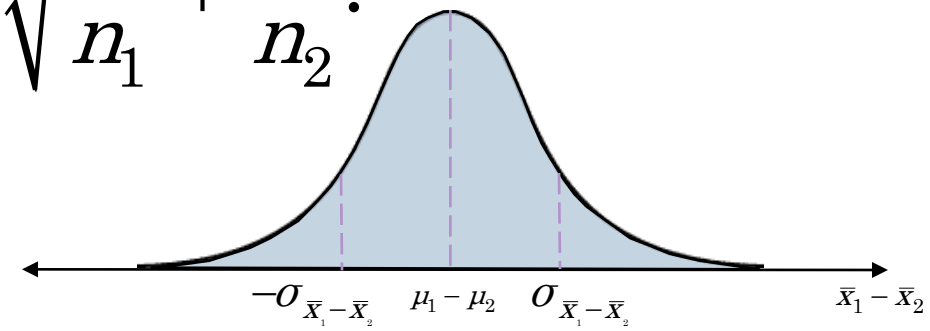
If these requirements are met, the sampling distribution for $\bar{x}_1 - \bar{x}_2$ (the difference of the sample means) is a normal distribution with mean and standard error of

$$\mu_{\bar{x}_1 - \bar{x}_2} = \mu_{\bar{x}_1} - \mu_{\bar{x}_2} = \mu_1 - \mu_2$$

and

$$\sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\sigma^2_{\bar{x}_1} + \sigma^2_{\bar{x}_2}} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}.$$

Sampling distribution for $\bar{x}_1 - \bar{x}_2$



Two Sample *z*-Test

Two-Sample z-Test for the Difference Between Means

A two-sample *z*-test can be used to test the difference between two population means μ_1 and μ_2 when a large sample (at least 30) is randomly selected from each population and the samples are independent. The test statistic is $\bar{x}_1 - \bar{x}_2$ and the standardized test statistic is

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sigma_{\bar{x}_1 - \bar{x}_2}} \quad \text{where} \quad \sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}.$$

When the samples are large, you can use s_1 and s_2 in place of σ_1 and σ_2. If the samples are not large, you can still use a two-sample z-test, provided the populations are normally distributed and the population standard deviations are known.



Two Sample *z*-Test for the Means

Using a Two-Sample z-Test for the Difference Between Means (Large Independent Samples)

1. State the claim mathematically. Identify the null and alternative hypotheses (State H0 and Ha)
2. Specify the level of significance. Identify α.
3. Sketch the sampling distribution. Determine the critical value(s). Determine the rejection regions(s).
4. Find the standardized test statistic. $$z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sigma_{\bar{x}_1 - \bar{x}_2}}$$
6. Make a decision to reject or fail to reject the null hypothesis.
7. Interpret the decision in the context of the original claim.
   If z is in the rejection region, reject H0. Otherwise, fail to reject H0.



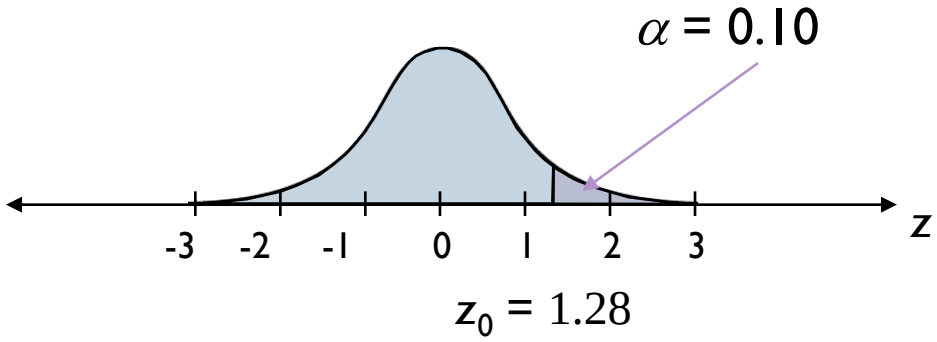
Two Sample *z*-Test for the Means

Example:

A high school math teacher claims that students in her class will score higher on the math portion of the ACT then students in a colleague's math class. The mean ACT math score for 49 students in her class is 22.1 and the standard deviation is 4.8. The mean ACT math score for 44 of the colleague's students is 19.8 and the standard deviation is 5.4. At $\alpha = 0.10$, can the teacher's claim be supported?

$H_0: \mu_1 \leq \mu_2$

$H_a: \mu_1 > \mu_2$ (Claim)

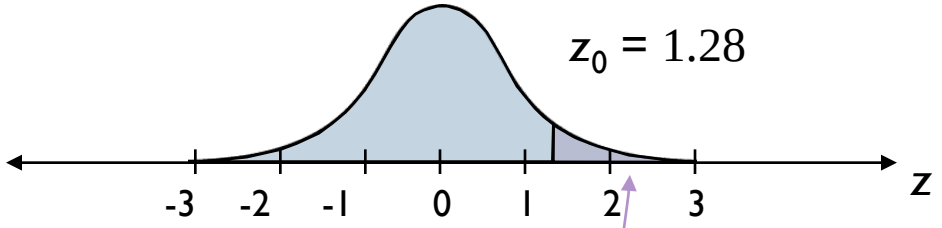


Two Sample *z*-Test for the Means

Example continued:

$H_0: \mu_1 \le \mu_2$

$H_a: \mu_1 > \mu_2$ (Claim)



Reject H_0.

The standardized error is

$$\sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} = \sqrt{\frac{4.8^2}{49} + \frac{5.4^2}{44}} \approx 1.0644.$$

The standardized test statistic is

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sigma_{\bar{x}_1 - \bar{x}_2}} = \frac{(22.1 - 19.8) - 0}{1.0644} \approx 2.161$$

There is enough evidence at the 10% level to support the teacher's claim that her students score better on the ACT.



Two Sample *t*-Test

If samples of size less than 30 are taken from normally-distributed populations, a *t*-test may be used to test the difference between the population means μ_1 and μ_2.

Three conditions are necessary to use a *t*-test for small independent samples.

1. The samples must be randomly selected.
2. The samples must be independent. Two samples are **independent** if the sample selected from one population is not related to the sample selected from the second population.
3. Each population must have a normal distribution.



Two Sample *t*-Test

Two-Sample *t*-Test for the Difference Between Means

A **two-sample *t*-test** is used to test the difference between two population means μ_1 and μ_2 when a sample is randomly selected from each population. Performing this test requires each population to be normally distributed, and the samples should be independent. The standardized test statistic is

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sigma_{\bar{x}_1 - \bar{x}_2}}.$$

If the population variances are equal, then information from the two samples is combined to calculate a **pooled estimate of the standard deviation** $\hat{\sigma}$.

$$\hat{\sigma} = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$



Two Sample *t*-Test

Two-Sample *t*-Test (Continued)

The standard error for the sampling distribution of $\bar{X}_1 - \bar{X}_2$ is

$$\sigma_{\bar{X}_1 - \bar{X}_2} = \hat{\sigma} \cdot \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$ Variances equal

and d.f.= $n_1 + n_2 - 2$.

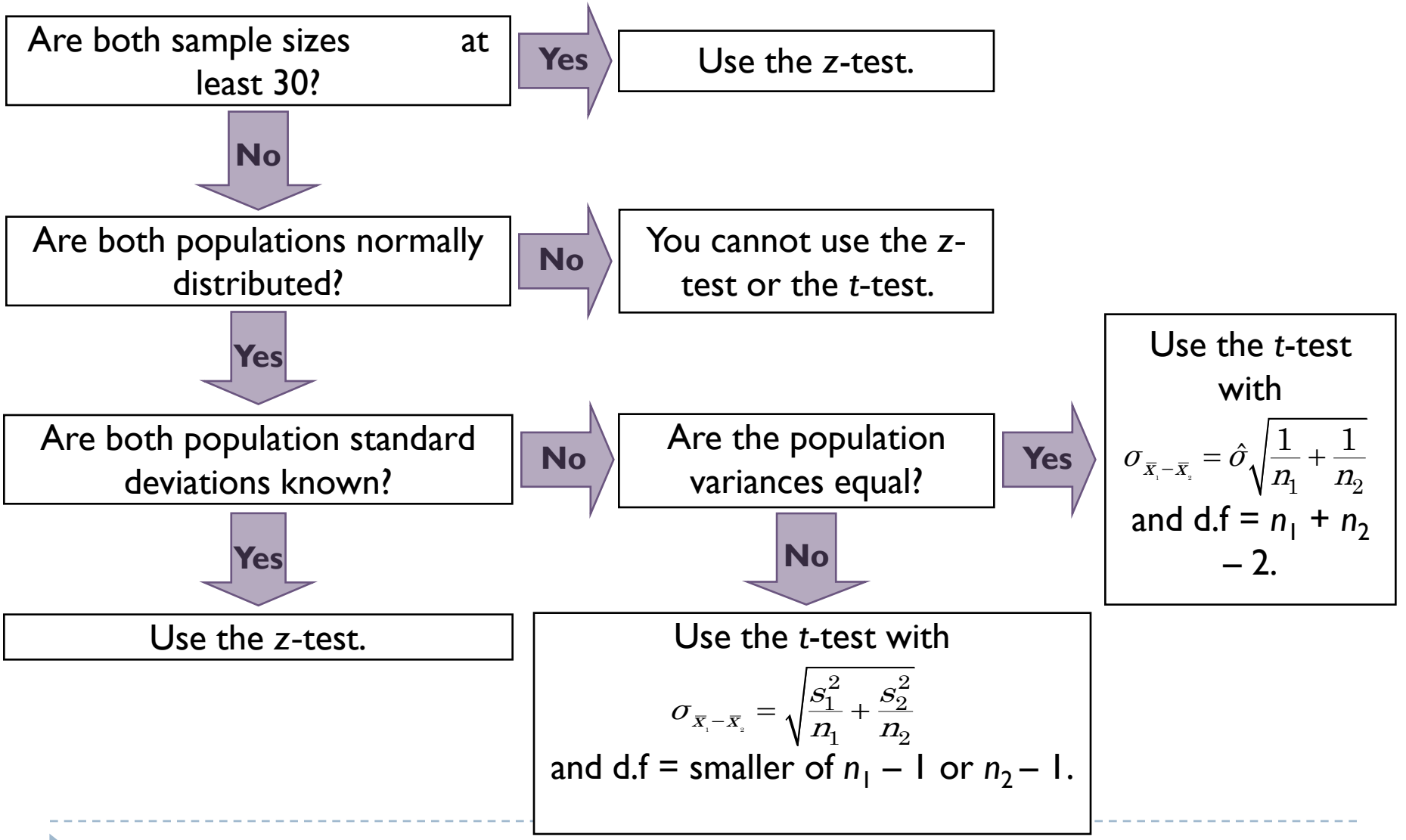
If the population variances are not equal, then the standard error is

$$\sigma_{\bar{X}_1 - \bar{X}_2} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$ Variances not equal

and d.f = smaller of $n_1 - 1$ or $n_2 - 1$.



Normal or *t*-Distribution?



Two Sample *t*-Test for the Means

Using a Two-Sample *t*-Test for the Difference Between Means (Small Independent Samples)

In Words	*In Symbols*
1. State the claim mathematically. Identify the null and alternative hypotheses.	State H_0 and H_a.
2. Specify the level of significance.	Identify α.
3. Identify the degrees of freedom and sketch the sampling distribution.	d.f. = $n_1 + n_2 - 2$ or d.f. = smaller of $n_1 - 1$ or $n_2 - 1$.
4. Determine the critical value(s).	



Two Sample *t*-Test for the Means

Using a Two-Sample *t*-Test for the Difference Between Means (Small Independent Samples)

In Words *In Symbols*

5. Determine the rejection regions(s).
6. Find the standardized test statistic.

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sigma_{\bar{x}_1 - \bar{x}_2}}$$

7. Make a decision to reject or fail to reject the null hypothesis.
8. Interpret the decision in the context of the original claim.

If t is in the rejection region, reject H_0. Otherwise, fail to reject H_0.



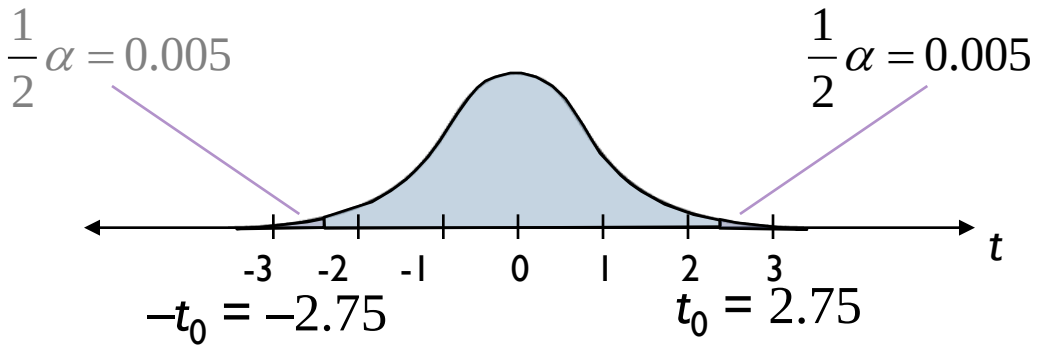
Two Sample *t*-Test for the Means

Example:

A random sample of 17 police officers in Brownsville has a mean annual income of $35,800 and a standard deviation of $7,800. In Greensville, a random sample of 18 police officers has a mean annual income of $35,100 and a standard deviation of $7,375. Test the claim at $\alpha = 0.01$ that the mean annual incomes in the two cities are not the same. Assume the population variances are equal.

$H_0: \mu_1 = \mu_2$

$H_a: \mu_1 \neq \mu_2$ (Claim)



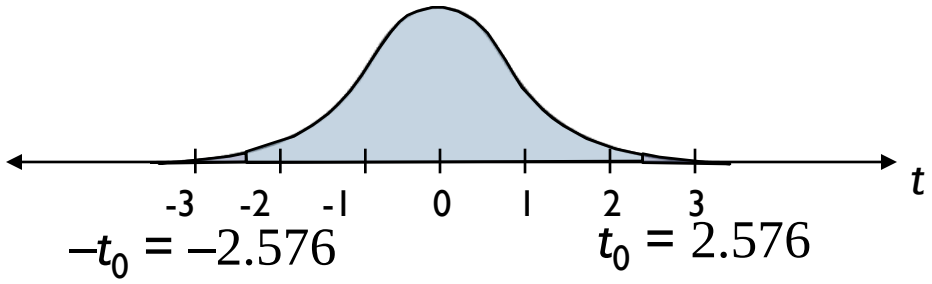
$$\text{d.f.} = n_1 + n_2 - 2$$
$$= 17 + 18 - 2 = 33$$

Two Sample *t*-Test for the Means

Example continued:

H_0: $\mu_1 = \mu_2$

H_a: $\mu_1 \neq \mu_2$ (Claim)



The standardized error is

$$\sigma_{\bar{x}_1 - \bar{x}_2} = \hat{\sigma}\sqrt{\frac{1}{n_1} + \frac{1}{n_2}} = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \cdot \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

$$= \sqrt{\frac{(17 - 1)7800^2 + (18 - 1)7375^2}{17 + 18 - 2}} \cdot \sqrt{\frac{1}{17} + \frac{1}{18}}$$

$$\approx 7584.0355(0.3382)$$

$$\approx 2564.92$$

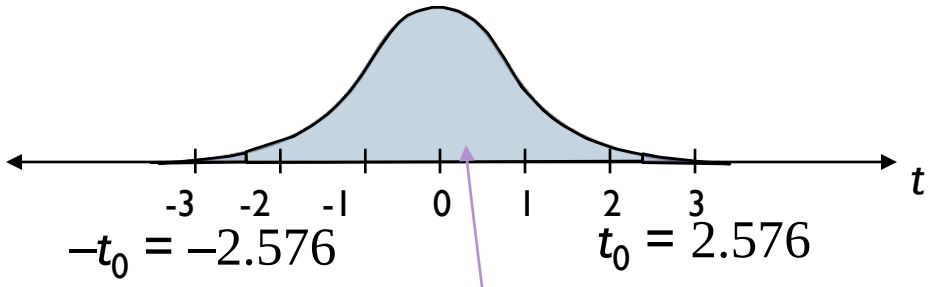


Two Sample *t*-Test for the Means

Example continued:

$H_0: \mu_1 = \mu_2$

$H_a: \mu_1 \neq \mu_2$ (Claim)



The standardized test statistic is

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sigma_{\bar{x}_1 - \bar{x}_2}} = \frac{(35800 - 35100) - 0}{2564.92} \approx 0.273$$

Fail to reject H_0.

There is not enough evidence at the 1% level to support the claim that the mean annual incomes differ.



t-Test for the Difference Between Means

Example:
A reading center claims that students will perform better on a standardized reading test after going through the reading course offered by their center. The table shows the reading scores of 6 students before and after the course. At $\alpha = 0.05$, is there enough evidence to conclude that the students' scores after the course are better than the scores before the course?

Student	1	2	3	4	5	6
Score (before)	85	96	70	76	81	78
Score (after)	88	85	89	86	92	89

$H_0: \mu_d \leq 0$

$H_a: \mu_d > 0$ (Claim)



t-Test for the Difference Between Means

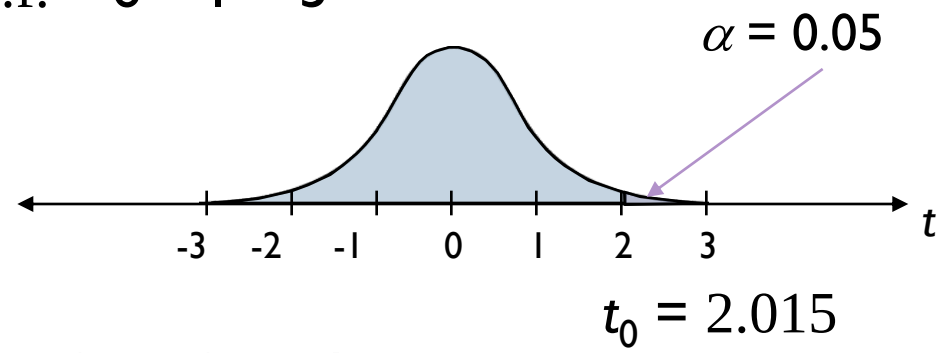
Example continued:

$H_0: \mu_d \leq 0$

$H_a: \mu_d > 0$ (Claim)

d = (score before) – (score after)

d.f. = 6 – 1 = 5



$t_0 = 2.015$

Student	1	2	3	4	5	6
Score (before)	85	96	70	76	81	78
Score (after)	88	85	89	86	92	89
d	–3	11	–19	–10	–11	–11
d^2	9	121	361	100	121	121

$\sum d = -43$

$\sum d^2 = 833$

$$\bar{d} = \frac{\sum d}{n} = \frac{-43}{6} \approx -7.167$$

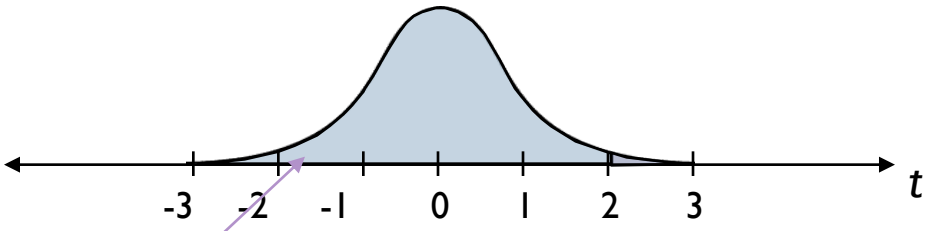
$$s_d = \sqrt{\frac{n(\sum d^2) - (\sum d)^2}{n(n-1)}} = \sqrt{\frac{6(833) - 1849}{6(5)}} \approx \sqrt{104.967} \approx 10.245$$

t-Test for the Difference Between Means

Example continued:

$H_0: \mu_d \leq 0$

$H_a: \mu_d > 0$ (Claim)



$t_0 = 2.015$

The standardized test statistic is

$$t = \frac{\bar{d} - \mu_d}{s_d/\sqrt{n}} = \frac{-7.167 - 0}{10.245/\sqrt{6}} \approx -1.714.$$

Fail to reject H_0.

There is not enough evidence at the 5% level to support the claim that the students' scores after the course are better than the scores before the course.



Two Sample z-Test for Proportions

A z-test is used to test the difference between two population proportions, p_1 and p_2.

Three conditions are required to conduct the test.

1. The samples must be randomly selected.
2. The samples must be independent.
3. The samples must be large enough to use a normal sampling distribution. That is,

$$n_1p_1 \geq 5,\ n_1q_1 \geq 5,$$

$$n_2p_2 \geq 5,\ \text{and}\ n_2q_2 \geq 5.$$



Two Sample z-Test for Proportions

If these conditions are met, then the sampling distribution for $\hat{p}_1 - \hat{p}_2$ is a normal distribution with mean

$$\mu_{\hat{p}_1 - \hat{p}_2} = p_1 - p_2$$

and standard error

$$\sigma_{\hat{p}_1 - \hat{p}_2} = \sqrt{\bar{p}\bar{q}\left(\frac{1}{n_1} + \frac{1}{n2_1}\right)}, \quad \text{where } \bar{q} = 1 - \bar{p}.$$

A weighted estimate of p_1 and p_2 can be found by using

$$\bar{p} = \frac{x_1 + x_2}{n_1 + n_2}, \text{ where } x_1 = n_1\hat{p}_1 \text{ and } x_2 = n_2\hat{p}_2.$$



Two Sample z-Test for Proportions

Two Sample *z*-Test for the Difference Between Proportions

A two sample *z*-test is used to test the difference between two population proportions p_1 and p_2 when a sample is randomly selected from each population.

The **test statistic** is

$$\hat{p}_1 - \hat{p}_1$$

and the **standardized test statistic** is

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\bar{p}\bar{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

where

$$\bar{p} = \frac{x_1 + x_2}{n_1 + n_2} \text{ and } \bar{q} = 1 - \bar{p}.$$

Note:

$n_1\bar{p}$, $n_1\bar{q}$, $n_2\bar{p}$, and $n_2\bar{q}$ must be at least 5.



Two Sample z-Test for Proportions

Using a Two-Sample *z*-Test for the Difference Between Proportions

	In Words	*In Symbols*
1.	State the claim. Identify the null and alternative hypotheses.	State H_0 and H_a.
2.	Specify the level of significance.	Identify α.
3.	Determine the critical value(s).	Use Table 4 in Appendix B.
4.	Determine the rejection region(s).	
5.	Find the weighted estimate of p_1 and p_2.	$\overline{p} = \frac{x_1 + x_2}{n_1 + n_2}$



Two Sample z-Test for Proportions

Using a Two-Sample *z*-Test for the Difference Between Proportions

	In Words	*In Symbols*
6.	Find the standardized test statistic.	$z = \dfrac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\overline{p}\,\overline{q}\left(\dfrac{1}{n_1} + \dfrac{1}{n_2}\right)}}$
7.	Make a decision to reject or fail to reject the null hypothesis.	If z is in the rejection region, reject H_0. Otherwise, fail to reject H_0.
8.	Interpret the decision in the context of the original claim.	

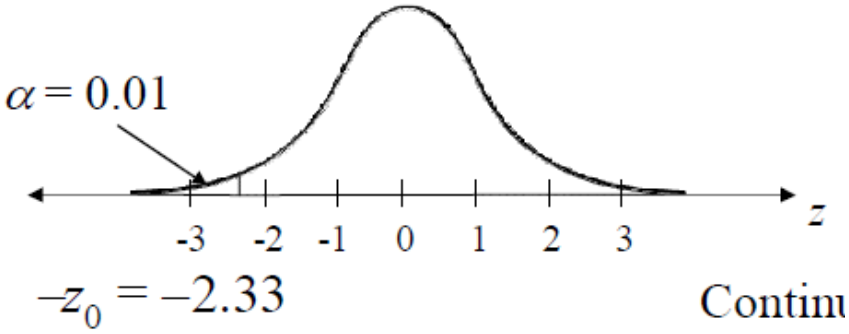


Two Sample z-Test for Proportions

Example:
A recent survey stated that male college students smoke less than female college students. In a survey of 1245 male students, 361 said they smoke at least one pack of cigarettes a day. In a survey of 1065 female students, 341 said they smoke at least one pack a day. At $\alpha = 0.01$, can you support the claim that the proportion of male college students who smoke at least one pack of cigarettes a day is lower then the proportion of female college students who smoke at least one pack a day?

$H_0: p_1 \geq p_2$

$H_a: p_1 < p_2$ (Claim)

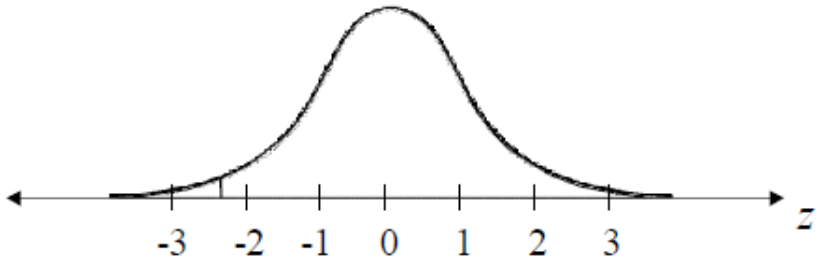


Two Sample z-Test for Proportions

Example continued:

$H_0: p_1 \geq p_2$

$H_a: p_1 < p_2$ (Claim)



$$\bar{p} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{n_1\hat{p}_1 + n_2\hat{p}_2}{n_1 + n_2} = \frac{361 + 341}{1245 + 1065} = \frac{702}{2310} \approx 0.304$$

$$\bar{q} = 1 - \bar{p} = 1 - 0.304 = 0.696$$

Because 1245(0.304), 1245(0.696), 1065(0.304), and 1065(0.696) are all at least 5, we can use a two-sample *z*-test.

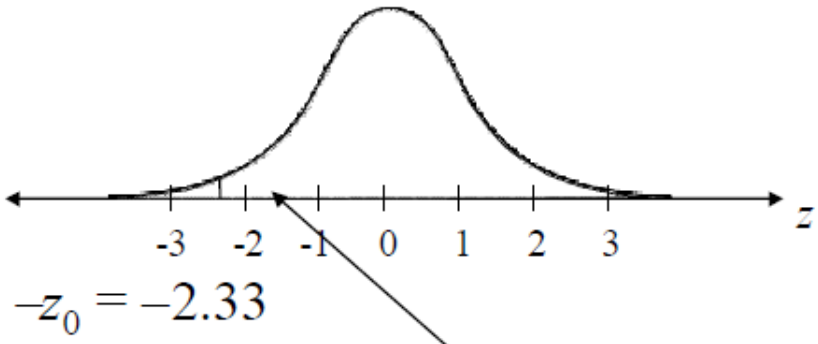


Two Sample z-Test for Proportions

Example continued:

$H_0: p_1 \geq p_2$

$H_a: p_1 < p_2$ (Claim)



$$z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\bar{p}\bar{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{(0.29 - 0.32) - 0}{\sqrt{(0.304)(0.696)\left(\frac{1}{1245} + \frac{1}{1065}\right)}} \approx -1.56$$

Fail to reject H_0.

There is not enough evidence at the 1% level to support the claim that the proportion of male college students who smoke is lower then the proportion of female college students who smoke.

